Criblage DFT (WB97X-D4/aug-cc-pVTZ) des allotropes polyazotés N$_x$: rapport d'étape

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Abstract

Ce document rend compte de la campagne DFT au niveau WB97X-D4/aug-cc-pVTZ sur les **193 candidats** présélectionnés par criblage GFN2-xTB (N_4 à N_{16}, familles neutre/cation/anion). **Les 193 jobs SLURM sont terminés** (0 erreur ORCA) ; parmi eux, **158 structures** (82 %) disposent d'un résultat DFT exploitable (géométrie optimisée et fréquences vérifiées), tandis que **35** (18 %) ont atteint la limite de cycles d'optimisation sans converger. Parmi ces 35, **28 sont en réalité déjà séparées en plusieurs composantes** à la dernière géométrie tentée (l'optimiseur suit un gradient de dissociation qui ne s'annule jamais en un nombre fini de cycles – probablement pas de minimum lié du tout pour ces topologies) ; seules **7** restent un cluster unique et sont des candidates crédibles à une reprise avec davantage de cycles (§4, Tableau S2). Un contrôle systématique de connectivité (§2) a par ailleurs révélé que 24 des 158 structures exploitables sont en réalité **fragmentées** (deux espèces moléculaires distinctes plutôt qu'un cluster lié, ex. N_2 + N_3^- – dont 11 déjà fragmentées dans la géométrie GFN2-xTB de départ, avant tout calcul DFT) – exclues des comparaisons de stabilité ci-dessous. Une confrontation aux bases de données et à la littérature déjà compilées dans `polyN-pipeline` montre que, parmi les 7 groupes (formule, charge) où une comparaison est possible, le criblage local a déjà trouvé un isomère non-fragmenté plus stable que tout ce qui est catalogué dans 3 d'entre eux (N_9^-, N_{10}, N_{12}), tandis que 165 des 193 candidats (86 %) ne correspondent à aucune topologie référencée ailleurs. Les deux ancres de référence pour les ΔH_f ioniques (N_5^- cyclique, N_5^+ coudé) ont été retrouvées dans `biblio_polyN/final_report_isodesmic.csv` et permettent de calculer un ΔH_f absolu (niveau GFN2-xTB) pour les 193 candidats – tables 2–4. Une seconde campagne (DLPNO-CCSD(T)-F12, 49 structures représentatives) est en cours pour établir des statistiques de temps de calcul et le nombre de cœurs optimal avant d'étendre le niveau CCSD(T) à l'ensemble des candidats.

Contents

1. Méthode

1.1 Origine des candidats

Les 193 structures de départ proviennent du criblage GFN2-xTB de `polyN-pipeline` (génération RDKit ETKDGv3 + énumération de graphes `geng`/nauty + substitution isolobale C→N + échantillonnage aléatoire, selon la formule), sélection par stoechiométrie et vérification de fréquences à ce niveau. Aucune contribution du second dépôt compagnon `polyN_adapt` (génération adaptative/surrogate) n'a été identifiée dans ce jeu – voir §2, à confirmer. Les formules à N impair (N_5, N_7, N_9, N_{11}, N_{13}, N_{15}) ne sont représentées qu'en cation/anion (contrainte de parité électronique), les formules paires principalement en neutre.

1.2 Méthodologies de génération des candidats (pools 1/2/3)

Trois pipelines distincts alimentent, ou peuvent alimenter, l'ensemble des candidats N_x^q considérés dans ce projet. Seul le **pool 1** est à l'origine des 193 candidats de cette campagne DFT (§1.1) ; les pools 2 et 3 sont décrits ici pour complétude méthodologique et parce qu'ils interviennent ailleurs dans ce rapport (comparaison de stabilité/topologie, §8).

Pool 1 – criblage combinatoire (`polyN-pipeline`), source de ce rapport
Génération de topologies candidates par trois voies combinées selon la formule : (i) énumération exhaustive de graphes par `geng`/nauty (degré borné), (ii) substitution isolobale C→N appliquée à des squelettes d'hydrocarbures C_xH_y récupérés sur PubChem (`harvest_hydrocarbons.py` → `cxhx_to_nx.py`), (iii) échantillonnage stochastique (`random_structure_generator.py`) pour les formules trop nombreuses pour l'énumération exhaustive. Chaque topologie est plongée en 3D (RDKit ETKDGv3), puis optimisée en deux temps au niveau GFN2-xTB, filtrée par vérification de fréquences (rejet des points-selles), et sélectionnée par critère énergétique (fenêtre au-dessus du meilleur isomère connu de la formule). Aucune contribution du pool 2 n'a été identifiée dans les 193 candidats de cette campagne (à confirmer, §2).

Pool 2 – surrogate adaptatif (`polyN_adapt`), non utilisé ici
Approche génération-par-génération : une population de graphes candidats (`geng` tant que l'énumération exhaustive reste praticable, puis mutation locale de l'archive accumulée – ajout/retrait d'arête, permutation double d'arêtes, à n constant – au-delà) est filtrée par un *surrogate* à deux volets : un régresseur à gradient boosté et une règle interprétable, tous deux réentraînés à chaque génération sur l'archive accumulée. Seuls les candidats

retenus par le surrogate sont réellement relaxés (GFN2-xTB via `tblite`, calculateur ASE in-process), avec calcul de fréquences pour confirmer un vrai minimum. Un apprentissage actif priorise l'évaluation réelle sur les candidats les plus incertains ou les plus proches de la frontière de la fenêtre d'énergie (0,2 eV/atome par défaut, recalculée dynamiquement par rapport au meilleur minimum connu) plutôt que sur tous les candidats acceptés indistinctement – objectif : minimiser le nombre de relaxations xTB réellement nécessaires face à un espace combinatoire trop vaste pour l'énumération exhaustive seule. Un raffinement post-xTB (Gaussian ou ORCA, chaine d'étapes configurable YAML, module `refinement/`) est prévu pour les quelques isomères bas en énergie qui survivent à la fenêtre. *Aucune bibliothèque de potentiel appris (type MACE) n'intervient comme surrogate dans le code actuel* de `polyN_adapt` – vérifié directement dans `surrogate/adaptive.py` ; seule une note de compatibilité OpenMP (`README.md`) mentionne `mace` comme bibliothèque tierce potentiellement importée par un script externe, sans lien avec le mécanisme de filtrage lui-même.

Pool 3 – structures issues de la littérature (`biblio_polyN`)
Structures extraites d'articles publiés (actuellement 2 sources complètement dépouillées, références [1] et [2] à l'Annexe A – §A.5) et de deux bases externes, `N_csp` et `polyN_study` – toutes relaxées localement au même niveau GFN2-xTB pour permettre une comparaison directe. C'est ce pool (337 structures catégorisées par formule et charge) qui sert de référentiel de comparaison en §8, indépendamment des candidats DFT de ce rapport. Une recherche de citations plus large (OpenAlex, 178 articles recensés dont 156 pas encore réduits en structures nommées) alimente ce même pool à terme – travail d'extraction restant listé à l'Annexe A (§A.5).

1.3 Niveau de calcul

Optimisation de géométrie puis fréquences, WB97X-D4/aug-cc-pVTZ, TightSCF, DEFGRID3 (ORCA 6.1.0). Deux protocoles selon la taille :

- **N_4–N_{11}** (« petits ») : un job unique Opt+Freq, sans contrainte de symétrie – évite tout risque de figer un candidat Jahn-Teller dans un point-selle de symétrie artificiellement élevée.
- **N_{12}–N_{16}** (« gros ») : chaîne en trois étapes Opt (sans symétrie) $\rightarrow$ ré-optimisation rapide avec `UseSym` (nettoie la géométrie vers le groupe ponctuel exact) $\rightarrow$ Freq avec `UseSym` (diagonalisation du Hessien exploitant la symétrie complète, pas seulement les sous-groupes de D_{2h}). Surcoût mesuré d'un facteur ~ 2 en temps de calcul sur un cas test (N_5^-, 7:41 $\rightarrow$ 15:35), jugé acceptable uniquement pour les grosses structures.

Distribution sur cluster SLURM via un ordonnanceur maison (`production_dispatch.py`) : surveillance en continu des cœurs libres sur les nœuds alloués, soumission automatique du prochain candidat dès qu'un créneau se libère, 8 cœurs par job (optimum mesuré : efficacité CPU 75–96 %, gain marginal au-delà ; 24–32 cœurs pour les gros tant que la file correspondante n'était pas épuisée, efficacité 90–96 % mesurée jusqu'à 32 cœurs sur N_{12}).

Tableau 1: Temps de calcul (écoulé = wall-clock, CPU = temps utilisateur+système cumulé sur tous les cœurs) sur les 193 jobs terminés au moment de la rédaction.

Catégorie	n	Wall moy. (min)	Wall min (min)	Wall max (min)	CPU-h moy.
Petits (8 cœurs, N4–N11)	166	90.5	5.1	289.1	11.05
Gros (24/32 cœurs, N12–N16)	27	140.2	23.3	548.6	54.94
Total	193	97.5	5.1	548.6	17.19

2. Provenance et statut de chaque candidat

C'est la table centrale de ce rapport : pour chacun des 193 candidats, son origine, son statut DFT actuel, son ΔH_f GFN2-xTB (ancré sur les valeurs littérature CCSD(T) retrouvées dans `biblio_polyN/final_report_isodesmic.csv` – $-106{,}4$ kcal/mol pour le pentagone N_5^- (D_{5h}), $195{,}9$ kcal/mol pour le N_5^+ coudé/Vchain (C_{2v}, valeur expérimentale par diffraction X concordante à $195{,}8$), $\Delta H_f(N_2) = 0$ pour les neutres par construction) et, quand une comparaison est possible, la topologie cataloguée à laquelle il correspond.

Les deux ancres ont été recalculées indépendamment ici (GFN2-xTB, xtb 6.7.1) et reproduisent les valeurs internes de `polyN_pipeline.py` à 5–6 décimales : $E(N_2) = -5{,}763935$ Ha, $E(N_5^-$, D_{5h}, = `N5_anion_anion_001`$) = -14{,}760899$ Ha, $E(N_5^+$, C_{2v}, = `N5_cation_cation_001`$) = -13{,}985267$ Ha. Point notable : la référence interne du pipeline pour la famille cation n'est pas ce dernier mais `N5_cation_cation_004` (isomère C_1 différent, $E = -13{,}816935$ Ha, $105{,}6$ kcal/mol moins stable) – les ΔH ci-dessous sont recalés sur le C_{2v} qui correspond à la structure littérature.

Trois tableaux distincts (neutres, cations, anions), triés par nombre d'atomes croissant puis par stabilité au sein de chaque formule ($\Delta H = 0$ pour l'isomère le plus stable, DFT quand disponible sinon GFN2-xTB). Chaque ligne étant un de nos 193 candidats, *our work* s'applique toujours ; quand la topologie correspond en plus à un article publié, son numéro est cité à côté ("`[k], our work`" – liste des articles en Annexe A). Sans correspondance article, seul *our work* apparaît – ce qui inclut, par choix explicite, les correspondances aux bases externes `N_csp` et `polyN_study` (candidats, pas des articles publiés) aussi bien que les topologies réellement inédites.

Portée de cette bibliographie (liste complète numérotée à l'Annexe A, §A.5) : les références [1]–[22] sont celles pour lesquelles `polyN-pipeline` a déjà extrait une structure et recalculé son ΔH_f isodesmique (110 structures nommées dans `table_isodesmic_{neutre,anion,cation}.tex`) – 8 de nos 193 candidats correspondent en plus à la topologie d'une de ces 22 références (priorité donnée à une correspondance `biblio_article` sur un doublon de base externe de même topologie mais moindre énergie). Les références [23]–[178] proviennent d'une recherche de citations plus large (OpenAlex, articles citant les 39 sources de `polyN-pipeline`) et n'ont **pas encore** de structure extraite : elles ne peuvent donc pas être citées par candidat tant que ce travail d'extraction n'est pas fait (liste du travail restant, Annexe A).

Tableau 2: Composés **neutres**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). N'inclut que les clusters liés (Tableau S1 : candidats fragmentés en plusieurs espèces moléculaires distinctes). Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	**N**	**PG**	**ΔH (kcal/mol)**	**niveau**	**Réf.**
`N4_neutral_neutral_001`	4	Td	0.00	DFT	our work
`N4_neutral_neutral_004`	4	D2h	3.38	DFT	our work
`N4_neutral_neutral_002`	4	C2v	16.85	DFT	our work
`N4_neutral_neutral_003`	4	C2v	33.07	DFT	our work
`N6_neutral_neutral_001`	6	C2	0.00	DFT	our work
`N6_neutral_neutral_002`	6	D2	23.07	DFT	[2], our work
`N6_neutral_neutral_003`	6	C1	28.07	DFT	our work
`N6_neutral_neutral_005`	6	C1	52.69	DFT	our work
`N6_neutral_neutral_007`	6	C2	53.36	DFT	our work
`N6_neutral_neutral_009`	6	C2	58.25	DFT	our work
`N6_neutral_neutral_008`	6	C2v	63.90	DFT	our work
`N6_neutral_neutral_006`	6	C1	85.49	DFT	our work
`N6_neutral_neutral_011`	6	–	119.57	xtb	our work

(suite)

N6_neutral_neutral_012	6	D3h	132.46	DFT	our work
N6_neutral_neutral_010	6	C2v	149.79	DFT	our work
N8_neutral_neutral_002	8	C1	0.00	DFT	[1],[2], our work
N8_neutral_neutral_001	8	D2h	11.23	DFT	[1],[2], our work
N8_neutral_neutral_005	8	C2	18.41	DFT	[2], our work
N8_neutral_neutral_003	8	C1	18.53	DFT	[2], our work
N8_neutral_neutral_008	8	C1	31.11	DFT	our work
N8_neutral_neutral_004	8	C1	36.64	DFT	[2], our work
N8_neutral_neutral_009	8	D3h	68.56	DFT	our work
N8_neutral_neutral_010	8	Cs	76.55	DFT	our work
N8_neutral_neutral_011	8	–	84.76	xtb	our work
N8_neutral_neutral_007	8	C2v	88.70	DFT	our work
N8_neutral_neutral_012	8	Cs	88.83	DFT	our work
N8_neutral_neutral_013	8	C2	101.43	DFT	our work
N8_neutral_neutral_019	8	D2d	114.22	DFT	our work
N8_neutral_neutral_020	8	C2h	118.58	DFT	our work
N8_neutral_neutral_016	8	Cs	122.30	DFT	our work
N8_neutral_neutral_015	8	–	126.26	xtb	our work
N8_neutral_neutral_014	8	C1	128.12	DFT	our work
N8_neutral_neutral_021	8	C2	129.00	DFT	our work
N8_neutral_neutral_017	8	–	139.04	xtb	our work
N8_neutral_neutral_022	8	C1	169.70	DFT	our work
N8_neutral_neutral_024	8	D2h	178.42	DFT	our work
N8_neutral_neutral_023	8	D2h	183.12	DFT	our work
N8_neutral_neutral_025	8	Oh	220.10	DFT	our work
N10_neutral_neutral_001	10	C2	0.00	DFT	our work
N10_neutral_neutral_002	10	C1	30.61	DFT	our work
N10_neutral_neutral_003	10	C1	50.60	DFT	[2], our work
N10_neutral_neutral_006	10	Cs	69.56	DFT	our work
N10_neutral_neutral_004	10	C1	71.13	DFT	[2], our work
N10_neutral_neutral_007	10	Cs	87.34	DFT	our work
N10_neutral_neutral_012	10	–	98.78	xtb	our work
N10_neutral_neutral_005	10	C1	102.77	DFT	our work
N10_neutral_neutral_011	10	C1	107.23	DFT	our work
N10_neutral_neutral_010	10	C1	118.99	DFT	our work
N10_neutral_neutral_015	10	C2	123.40	DFT	our work
N10_neutral_neutral_016	10	C1	126.63	DFT	our work
N10_neutral_neutral_014	10	C1	139.29	DFT	our work
N10_neutral_neutral_017	10	C2v	151.14	DFT	our work
N10_neutral_neutral_021	10	C2v	173.84	DFT	our work
N10_neutral_neutral_013	10	C2v	193.75	DFT	our work
N10_neutral_neutral_018	10	C2	199.07	DFT	our work
N10_neutral_neutral_019	10	C3v	203.80	DFT	our work
N10_neutral_neutral_020	10	C2	214.21	DFT	our work
N10_neutral_neutral_022	10	D5h	241.14	DFT	our work
N12_neutral_neutral_001	12	C1	0.00	DFT	our work
N12_neutral_neutral_003	12	C1	36.41	DFT	our work
N12_neutral_neutral_002	12	C1	56.99	DFT	our work
N12_neutral_neutral_004	12	Cs	116.12	DFT	our work
N12_neutral_neutral_006	12	C1	127.50	DFT	our work
N12_neutral_neutral_005	12	C1	138.12	DFT	our work
N12_neutral_neutral_007	12	D3d	188.33	DFT	our work
N12_neutral_neutral_008	12	C1	219.30	DFT	our work
N12_neutral_neutral_009	12	D6h	334.92	DFT	our work
N14_neutral_neutral_002	14	C1	0.00	DFT	our work
N14_neutral_neutral_001	14	C1	4.42	DFT	our work
N14_neutral_neutral_003	14	C2v	22.06	DFT	our work
N14_neutral_neutral_005	14	C2	95.19	DFT	our work
N16_neutral_neutral_002	16	C1	0.00	DFT	our work
N16_neutral_neutral_001	16	C1	40.23	DFT	our work

Note : ΔH = énergie d'isomérisation à N et charge fixés, N_x^q(isomère i) $\rightarrow$ N_x^q(isomère le plus stable de la même formule), $\Delta H = E(i) - E_{\mathrm{stable}}$, au niveau DFT quand disponible sinon GFN2-xTB (colonne *niveau*).

Tableau 3: Composés **cationiques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). N'inclut que les clusters liés (Tableau S1 : candidats fragmentés en plusieurs espèces moléculaires distinctes). Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	**N**	**PG**	**ΔH (kcal/mol)**	**niveau**	**Réf.**
N5_cation_cation_003	5	C1	0.00	DFT	our work
N5_cation_cation_001	5	C2v	0.00	DFT	our work
N5_cation_cation_004	5	C1	0.01	DFT	our work
N5_cation_cation_002	5	Cs	41.63	DFT	our work
N5_cation_cation_005	5	D2d	109.53	DFT	our work
N5_cation_cation_006	5	C4v	155.74	DFT	our work
N7_cation_cation_001	7	Cs	0.00	DFT	our work
N7_cation_cation_002	7	–	6.20	xtb	our work
N7_cation_cation_003	7	Cs	25.42	DFT	our work
N7_cation_cation_004	7	–	91.26	xtb	our work
N7_cation_cation_005	7	Cs	189.69	DFT	our work
N7_cation_cation_006	7	C1	200.56	DFT	our work
N9_cation_cation_002	9	–	0.00	xtb	our work
N9_cation_cation_006	9	C2v	0.00	DFT	our work
N9_cation_cation_003	9	–	2.53	xtb	our work
N9_cation_cation_004	9	–	4.76	xtb	our work
N9_cation_cation_008	9	C1	7.40	DFT	our work
N9_cation_cation_009	9	C1	38.07	DFT	our work
N9_cation_cation_012	9	C2v	57.61	DFT	our work
N9_cation_cation_005	9	–	71.16	xtb	our work
N9_cation_cation_011	9	Cs	82.84	DFT	our work
N9_cation_cation_007	9	–	84.43	xtb	our work
N9_cation_cation_010	9	–	102.87	xtb	our work
N11_cation_cation_002	11	C1	-23.30	DFT	our work
N11_cation_cation_001	11	C2v	0.00	DFT	our work
N11_cation_cation_005	11	C2	22.54	DFT	our work
N11_cation_cation_004	11	C1	32.75	DFT	our work
N11_cation_cation_006	11	C1	44.38	DFT	our work
N11_cation_cation_003	11	–	46.08	xtb	our work
N11_cation_cation_007	11	C1	95.39	DFT	our work
N11_cation_cation_008	11	–	195.40	xtb	our work
N12_cation_cation_001	12	C1	0.00	DFT	our work
N13_cation_cation_001	13	C1	0.00	DFT	our work
N13_cation_cation_002	13	–	40.46	xtb	our work
N13_cation_cation_003	13	–	56.53	xtb	our work
N13_cation_cation_004	13	C1	151.22	DFT	our work
N15_cation_cation_001	15	C1	0.00	DFT	our work
N15_cation_cation_002	15	C1	327.87	DFT	our work

Note : ΔH = énergie d'isomérisation à N et charge fixés, N_x^q(isomère i) $\rightarrow$ N_x^q(isomère le plus stable de la même formule), $\Delta H = E(i) - E_{\mathrm{stable}}$, au niveau DFT quand disponible sinon GFN2-xTB (colonne *niveau*).

Tableau 4: Composés **anioniques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). N'inclut que les clusters liés (Tableau S1 : candidats fragmentés en plusieurs espèces moléculaires distinctes). Référence(s) : numéro(s) d'article (Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	**N**	**PG**	**ΔH (kcal/mol)**	**niveau**	**Réf.**
N5_anion_anion_001	5	D5h	0.00	DFT	our work
N5_anion_anion_006	5	Cs	104.46	DFT	our work
N5_anion_anion_010	5	C2v	170.76	DFT	our work
N7_anion_anion_002	7	C1	0.00	DFT	our work
N7_anion_anion_001	7	C1	0.00	DFT	our work
N7_anion_anion_005	7	C1	2.22	DFT	our work

(suite)

N7_anion_anion_004	7	C1	7.80	DFT	our work
N7_anion_anion_013	7	C1	9.42	DFT	our work
N7_anion_anion_003	7	C2v	20.91	DFT	our work
N7_anion_anion_011	7	C1	26.27	DFT	our work
N7_anion_anion_012	7	C1	29.35	DFT	our work
N7_anion_anion_007	7	Cs	35.76	DFT	our work
N7_anion_anion_016	7	C1	40.69	DFT	our work
N7_anion_anion_010	7	C2v	46.25	DFT	our work
N7_anion_anion_008	7	C1	49.34	DFT	our work
N7_anion_anion_009	7	C1	54.96	DFT	our work
N7_anion_anion_014	7	Cs	55.87	DFT	our work
N7_anion_anion_015	7	–	71.61	xtb	our work
N7_anion_anion_018	7	Cs	79.33	DFT	our work
N7_anion_anion_017	7	Cs	80.80	DFT	our work
N7_anion_anion_019	7	C1	89.71	DFT	our work
N7_anion_anion_020	7	C1	179.57	DFT	our work
N9_anion_anion_001	9	C1	0.00	DFT	our work
N9_anion_anion_002	9	–	19.37	xtb	our work
N9_anion_anion_006	9	Cs	27.56	DFT	our work
N9_anion_anion_003	9	C1	40.85	DFT	our work
N9_anion_anion_008	9	C1	48.03	DFT	our work
N9_anion_anion_004	9	C1	50.71	DFT	our work
N9_anion_anion_011	9	Cs	58.34	DFT	our work
N9_anion_anion_005	9	C2	59.03	DFT	our work
N9_anion_anion_010	9	C1	62.59	DFT	our work
N9_anion_anion_012	9	C1	62.73	DFT	our work
N9_anion_anion_014	9	–	88.41	xtb	our work
N9_anion_anion_013	9	C1	96.48	DFT	our work
N9_anion_anion_018	9	C1	98.82	DFT	our work
N9_anion_anion_017	9	–	128.05	xtb	our work
N9_anion_anion_019	9	C1	139.31	DFT	our work
N9_anion_anion_016	9	C2v	164.79	DFT	our work
N11_anion_anion_001	11	C1	-60.32	DFT	our work
N11_anion_anion_002	11	C1	-21.85	DFT	our work
N11_anion_anion_006	11	C1	0.00	DFT	our work
N11_anion_anion_005	11	C1	14.07	DFT	our work
N11_anion_anion_009	11	–	21.88	xtb	our work
N11_anion_anion_004	11	C1	31.83	DFT	our work
N11_anion_anion_007	11	Cs	41.41	DFT	our work
N11_anion_anion_010	11	Cs	41.94	DFT	our work
N11_anion_anion_008	11	C1	48.28	DFT	our work
N11_anion_anion_003	11	C2	48.86	DFT	our work
N11_anion_anion_012	11	–	51.30	xtb	our work
N11_anion_anion_014	11	–	62.28	xtb	our work
N11_anion_anion_015	11	–	88.62	xtb	our work
N11_anion_anion_016	11	–	106.60	xtb	our work
N11_anion_anion_013	11	C1	116.68	DFT	our work
N11_anion_anion_017	11	C1	125.14	DFT	our work
N13_anion_anion_001	13	C1	0.00	DFT	our work
N13_anion_anion_002	13	–	115.00	xtb	our work
N13_anion_anion_003	13	–	116.61	xtb	our work
N13_anion_anion_004	13	–	123.73	xtb	our work

Note : ΔH = énergie d'isomérisation à N et charge fixés, N_x^q(isomère i) $\rightarrow$ N_x^q(isomère le plus stable de la même formule), $\Delta H = E(i) - E_{\text{stable}}$, au niveau DFT quand disponible sinon GFN2-xTB (colonne *niveau*).

3. Classement de stabilité : DFT contre xTB

Les tableaux suivants classent chaque isomère par rang DFT croissant (rang 1 = ground state DFT) au sein de sa formule, avec le rang xTB d'origine rappelé en regard – **la question posée est de savoir si le criblage xTB, nettement moins cher, prédit le même classement de stabilité que le DFT**, ou si des réordonnancements significatifs apparaissent.

Tableau 5: Classement DFT vs xTB, composés **neutres**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N4_neutral_neutral_001	4	1	1	0.0	0.0	✓
N4_neutral_neutral_004	4	4	2	41.64	3.375	✓
N4_neutral_neutral_002	4	2	3	26.47	16.846	✓
N4_neutral_neutral_003	4	3	4	31.01	33.074	✓
N6_neutral_neutral_001	6	1	1	0.0	0.0	✓
N6_neutral_neutral_002	6	2	2	24.09	23.072	✓
N6_neutral_neutral_003	6	3	3	55.0	28.065	✓
N6_neutral_neutral_005	6	5	4	64.62	52.687	✓
N6_neutral_neutral_007	6	7	5	83.63	53.362	✓
N6_neutral_neutral_009	6	9	6	110.8	58.246	✓
N6_neutral_neutral_008	6	8	7	84.52	63.9	✓
N6_neutral_neutral_006	6	6	8	78.8	85.49	✓
N6_neutral_neutral_012	6	12	9	136.66	132.463	✓
N6_neutral_neutral_010	6	10	10	114.33	149.792	✓
N8_neutral_neutral_002	8	2	1	9.91	0.0	
N8_neutral_neutral_001	8	1	2	0.0	11.226	
N8_neutral_neutral_005	8	5	3	29.41	18.413	✓
N8_neutral_neutral_003	8	3	4	20.26	18.532	✓
N8_neutral_neutral_008	8	8	5	67.22	31.111	✓
N8_neutral_neutral_004	8	4	6	28.09	36.638	✓
N8_neutral_neutral_009	8	9	7	73.7	68.563	✓
N8_neutral_neutral_010	8	10	8	84.06	76.549	✓
N8_neutral_neutral_007	8	7	9	66.38	88.7	✓
N8_neutral_neutral_012	8	12	10	93.97	88.827	✓
N8_neutral_neutral_013	8	13	11	102.9	101.433	✓
N8_neutral_neutral_019	8	19	12	150.3	114.216	✓
N8_neutral_neutral_020	8	20	13	152.93	118.581	✓
N8_neutral_neutral_016	8	16	14	132.78	122.296	✓
N8_neutral_neutral_014	8	14	15	113.23	128.118	✓
N8_neutral_neutral_021	8	21	16	155.66	129.004	✓
N8_neutral_neutral_022	8	22	17	161.85	169.696	✓
N8_neutral_neutral_024	8	24	18	183.05	178.416	✓
N8_neutral_neutral_023	8	23	19	164.68	183.123	✓
N8_neutral_neutral_025	8	25	20	205.7	220.095	✓
N10_neutral_neutral_001	10	1	1	0.0	0.0	✓
N10_neutral_neutral_002	10	2	2	10.5	30.609	✓
N10_neutral_neutral_003	10	3	3	23.5	50.598	✓
N10_neutral_neutral_006	10	6	4	38.56	69.561	✓
N10_neutral_neutral_004	10	4	5	33.36	71.134	✓
N10_neutral_neutral_007	10	7	6	53.65	87.344	✓
N10_neutral_neutral_005	10	5	7	36.56	102.766	✓
N10_neutral_neutral_011	10	11	8	92.95	107.23	✓
N10_neutral_neutral_010	10	10	9	88.62	118.993	✓
N10_neutral_neutral_015	10	15	10	111.24	123.395	✓
N10_neutral_neutral_016	10	16	11	113.83	126.633	✓
N10_neutral_neutral_014	10	14	12	100.5	139.293	✓
N10_neutral_neutral_017	10	17	13	130.46	151.139	✓
N10_neutral_neutral_021	10	21	14	174.6	173.835	✓
N10_neutral_neutral_013	10	13	15	99.55	193.748	✓
N10_neutral_neutral_018	10	18	16	143.02	199.068	✓
N10_neutral_neutral_019	10	19	17	147.97	203.805	✓
N10_neutral_neutral_020	10	20	18	168.46	214.208	✓
N10_neutral_neutral_022	10	22	19	183.94	241.136	✓
N12_neutral_neutral_001	12	1	1	0.0	0.0	✓
N12_neutral_neutral_003	12	3	2	35.91	36.408	✓
N12_neutral_neutral_002	12	2	3	13.24	56.987	✓
N12_neutral_neutral_004	12	4	4	45.58	116.12	✓
N12_neutral_neutral_006	12	6	5	76.36	127.5	✓
N12_neutral_neutral_005	12	5	6	68.53	138.121	✓
N12_neutral_neutral_007	12	7	7	100.96	188.328	✓
N12_neutral_neutral_008	12	8	8	133.16	219.304	✓
N12_neutral_neutral_009	12	9	9	242.43	334.919	✓

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N14_neutral_neutral_002	14	2	1	40.76	0.0	
N14_neutral_neutral_001	14	1	2	0.0	4.425	
N14_neutral_neutral_003	14	3	3	77.89	22.063	✓
N14_neutral_neutral_005	14	5	4	133.34	95.193	✓
N16_neutral_neutral_002	16	2	1	36.7	0.0	
N16_neutral_neutral_001	16	1	2	0.0	40.228	

Accord de rang xTB/DFT par groupe (structure classée au même rang par les deux méthodes, sur l'effectif total du groupe) : N4: 1/4, N6: 4/10, N8: 0/20, N10: 3/19, N12: 5/9, N14: 1/4, N16: 0/2 – soit 14/68 structures au total pour les neutres.

Tableau 6: Classement DFT vs xTB, composés **cationiques**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N5_cation_cation_003	5	3	1	104.85	0.0	
N5_cation_cation_001	5	1	2	0.0	0.001	
N5_cation_cation_004	5	4	3	105.63	0.007	✓
N5_cation_cation_002	5	2	4	74.16	41.634	✓
N5_cation_cation_005	5	5	5	134.0	109.535	✓
N5_cation_cation_006	5	6	6	184.33	155.738	✓
N7_cation_cation_001	7	1	1	0.0	0.0	✓
N7_cation_cation_003	7	3	2	27.01	25.419	✓
N7_cation_cation_005	7	5	3	124.94	189.693	✓
N7_cation_cation_006	7	6	4	178.83	200.561	✓
N9_cation_cation_006	9	6	1	0.0	0.0	✓
N9_cation_cation_008	9	8	2	10.35	7.403	✓
N9_cation_cation_009	9	9	3	12.81	38.071	✓
N9_cation_cation_012	9	12	4	72.31	57.612	✓
N9_cation_cation_011	9	11	5	69.62	82.84	✓
N11_cation_cation_002	11	2	1	2.67	0.0	
N11_cation_cation_001	11	1	2	0.0	23.301	
N11_cation_cation_005	11	5	3	73.62	45.845	✓
N11_cation_cation_004	11	4	4	61.37	56.05	✓
N11_cation_cation_006	11	6	5	100.27	67.685	✓
N11_cation_cation_007	11	7	6	128.07	118.687	✓
N12_cation_cation_001	12	1	1	0.0	0.0	✓
N13_cation_cation_001	13	1	1	0.0	0.0	✓
N13_cation_cation_004	13	4	2	194.1	151.221	✓
N15_cation_cation_001	15	1	1	0.0	0.0	✓
N15_cation_cation_002	15	2	2	259.27	327.866	✓

Accord de rang xTB/DFT par groupe (structure classée au même rang par les deux méthodes, sur l'effectif total du groupe) : N5: 2/6, N7: 1/4, N9: 0/5, N11: 1/6, N12: 1/1, N13: 1/2, N15: 2/2 – soit 8/26 structures au total pour les cationiques.

Tableau 7: Classement DFT vs xTB, composés **anioniques**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N5_anion_anion_001	5	1	1	0.0	0.0	✓
N5_anion_anion_006	5	6	2	104.25	104.46	✓
N5_anion_anion_010	5	10	3	143.36	170.763	✓

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N7_anion_anion_002	7	2	1	2.77	0.0	
N7_anion_anion_001	7	1	2	0.0	0.001	
N7_anion_anion_005	7	5	3	22.23	2.22	✓
N7_anion_anion_004	7	4	4	9.12	7.804	✓
N7_anion_anion_013	7	13	5	54.11	9.422	✓
N7_anion_anion_003	7	3	6	7.94	20.912	✓
N7_anion_anion_011	7	11	7	51.7	26.266	✓
N7_anion_anion_012	7	12	8	52.34	29.348	✓
N7_anion_anion_007	7	7	9	41.65	35.755	✓
N7_anion_anion_016	7	16	10	74.29	40.691	✓
N7_anion_anion_010	7	10	11	50.4	46.247	✓
N7_anion_anion_008	7	8	12	46.21	49.337	✓
N7_anion_anion_009	7	9	13	47.46	54.955	✓
N7_anion_anion_014	7	14	14	59.93	55.871	✓
N7_anion_anion_018	7	18	15	96.97	79.33	✓
N7_anion_anion_017	7	17	16	82.01	80.797	✓
N7_anion_anion_019	7	19	17	102.35	89.71	✓
N7_anion_anion_020	7	20	18	144.93	179.573	✓
N9_anion_anion_001	9	1	1	0.0	0.0	✓
N9_anion_anion_006	9	6	2	35.95	27.555	✓
N9_anion_anion_003	9	3	3	23.62	40.849	✓
N9_anion_anion_008	9	8	4	53.91	48.032	✓
N9_anion_anion_004	9	4	5	29.47	50.709	✓
N9_anion_anion_011	9	11	6	60.0	58.339	✓
N9_anion_anion_005	9	5	7	33.08	59.031	✓
N9_anion_anion_010	9	10	8	57.07	62.594	✓
N9_anion_anion_012	9	12	9	86.79	62.733	✓
N9_anion_anion_013	9	13	10	87.4	96.479	✓
N9_anion_anion_018	9	18	11	141.31	98.819	✓
N9_anion_anion_019	9	19	12	153.96	139.31	✓
N9_anion_anion_016	9	16	13	126.07	164.787	✓
N11_anion_anion_001	11	1	1	0.0	0.0	✓
N11_anion_anion_002	11	2	2	39.44	38.478	✓
N11_anion_anion_006	11	6	3	57.42	60.325	✓
N11_anion_anion_005	11	5	4	51.65	74.395	✓
N11_anion_anion_004	11	4	5	49.83	92.156	✓
N11_anion_anion_007	11	7	6	69.82	101.735	✓
N11_anion_anion_010	11	10	7	89.5	102.267	✓
N11_anion_anion_008	11	8	8	77.42	108.607	✓
N11_anion_anion_003	11	3	9	44.22	109.185	✓
N11_anion_anion_013	11	13	10	115.91	177.005	✓
N11_anion_anion_017	11	17	11	164.9	185.468	✓
N13_anion_anion_001	13	1	1	0.0	0.0	✓

Accord de rang xTB/DFT par groupe (structure classée au même rang par les deux méthodes, sur l'effectif total du groupe) : N5: 1/3, N7: 2/18, N9: 2/13, N11: 3/11, N13: 1/1 – soit 9/46 structures au total pour les anioniques.

4. Vérification des minima

Les 193 jobs SLURM de cette campagne sont tous terminés, mais **35 d'entre eux (18 %) n'ont pas convergé** – l'optimiseur DFT a atteint sa limite de cycles (`The optimization did not converge but reached the maximum`) sans trouver de point stationnaire ; leur énergie n'est donc pas exploitable en l'état. Un contrôle de connectivité sur la dernière géométrie tentée (et non la géométrie xTB de départ) montre que **28 des 35 sont déjà séparées en plusieurs composantes connexes** à ce dernier cycle – l'optimiseur suit une dissociation en cours plutôt qu'un vrai problème numérique de convergence, et une reprise avec plus de cycles ne trouvera vraisemblablement aucun minimum lié pour ces topologies. Seules les **7** restantes (cluster unique au dernier cycle) sont des candidates crédibles à une reprise. Liste complète avec le nombre de cycles atteint, le statut de fragmentation au dernier cycle et représentation avant (géométrie xTB de départ) / dernier cycle tenté : Tableau S2 et Figure S2 (Annexe, §A.2). Ces 35 sont exclues de tous les tableaux d'analyse ci-dessous et devront être reprises avec davantage de cycles d'optimisation (`%geom MaxIter`) dans une prochaine passe. Les **158 structures restantes** disposent d'un bloc de thermochimie complet, parmi lesquelles **38 présentent au moins une fréquence imaginaire** et ne sont donc pas, en l'état, de vrais minima locaux. Contrairement au protocole gros-cluster, aucune redésymétrisation automatique n'est appliquée : chaque candidat flaggé devra être repris manuellement (distorsion le long du mode imaginaire puis réoptimisation) avant d'être retenu comme candidat final – exactement la logique de la colonne `is_true_minimum` / `n_freq_hops` déjà implémentée dans `polyN_pipeline.py` au niveau GFN2-xTB, appliquée ici au niveau DFT. Liste complète avec la fréquence imaginaire dominante (la plus négative), son intensité IR et l'énergie relative au ground-state DFT de la même composition : Tableau S2 (Annexe) – cette dernière colonne permet de vérifier si l'instabilité vibrationnelle concentre sur les isomères déjà très haut en énergie (attendu) ou touche aussi des candidats proches du ground-state (auquel cas ce serait plus préoccupant pour le classement de stabilité).

Un contrôle séparé de connectivité (composantes connexes du graphe N–N à 1,70 Å) identifie par ailleurs **24 candidats fragmentés** – deux ou plusieurs espèces moléculaires distinctes (ex. N_2 + N_3^-) plutôt qu'un cluster N_x lié unique, listés à part en Tableau S1 avec représentation avant/après en Figure S1. Ce contrôle a été appliqué également à la géométrie de départ (avant tout calcul DFT) pour les 24 : **11 étaient déjà fragmentées dans la géométrie GFN2-xTB issue du criblage**, avant même d'entrer dans cette campagne DFT (artefact du criblage en amont, colonne *déjà frag. à xTB* du Tableau S1) ; les 13 autres étaient des clusters intacts à xTB et n'ont fragmenté que pendant l'optimisation DFT. Les tableaux d'analyse (§2–7) et la comparaison de stabilité (§8) portent uniquement sur les clusters liés : une espèce dissociée est souvent plus basse en énergie qu'un vrai cluster lié, ce qui fausserait tout classement fondé sur l'énergie seule.

5. Orbitales frontières

Tableau 8: Orbitales frontières (WB97X-D4/aug-cc-pVTZ) des candidats terminés.

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N4_neutral_neutral_001	Td	-13.9405	1.7935	15.7340
N4_neutral_neutral_004	D2h	-10.6441	-2.3507	8.2934
N4_neutral_neutral_002	C2v	-9.2289	-1.2933	7.9356
N4_neutral_neutral_003	C2v	-10.5771	-1.9734	8.6037
N5_anion_anion_001	D5h	-5.8838	5.8881	11.7719
N5_anion_anion_006	Cs	-1.8642	3.7995	5.6637
N5_anion_anion_010	C2v	-1.9078	5.4320	7.3398
N5_cation_cation_003	C1	-18.4487	-6.5897	11.8590
N5_cation_cation_001	C2v	-18.4486	-6.5903	11.8583
N5_cation_cation_004	C1	-18.4488	-6.5881	11.8607
N5_cation_cation_002	Cs	-19.4731	-7.2442	12.2289
N5_cation_cation_005	D2d	-19.7750	-8.2030	11.5720
N5_cation_cation_006	C4v	-20.6577	-7.9402	12.7175
N6_neutral_neutral_001	C2	-9.2804	-0.0404	9.2400
N6_neutral_neutral_002	D2	-11.4121	-2.0522	9.3599
N6_neutral_neutral_003	C1	-10.1854	-0.1470	10.0384
N6_neutral_neutral_005	C1	-10.0047	-1.3166	8.6881
N6_neutral_neutral_007	C2	-12.1087	-0.8535	11.2552
N6_neutral_neutral_009	C2	-11.2453	-0.2179	11.0274
N6_neutral_neutral_008	C2v	-12.0044	-0.7828	11.2216
N6_neutral_neutral_006	C1	-9.3271	-0.6904	8.6367
N6_neutral_neutral_012	D3h	-11.9779	0.1531	12.1310
N6_neutral_neutral_010	C2v	-9.7889	-3.6564	6.1325
N7_anion_anion_002	C1	-1.9591	4.7110	6.6701

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N7_anion_anion_001	C1	-1.9595	4.7108	6.6703
N7_anion_anion_005	C1	-3.4999	4.4963	7.9962
N7_anion_anion_004	C1	-4.0142	4.5633	8.5775
N7_anion_anion_013	C1	-3.7378	4.5618	8.2996
N7_anion_anion_003	C2v	-3.1498	5.2806	8.4304
N7_anion_anion_011	C1	-2.2849	4.8177	7.1026
N7_anion_anion_012	C1	-2.2126	4.8717	7.0843
N7_anion_anion_007	Cs	-2.2941	4.1099	6.4040
N7_anion_anion_016	C1	-3.5935	4.2032	7.7967
N7_anion_anion_010	C2v	-4.2100	5.1843	9.3943
N7_anion_anion_008	C1	-2.9901	4.4265	7.4166
N7_anion_anion_009	C1	-3.0117	4.5488	7.5605
N7_anion_anion_014	Cs	-3.7877	5.0235	8.8112
N7_anion_anion_018	Cs	-2.9551	4.2456	7.2007
N7_anion_anion_017	Cs	-4.3231	4.7093	9.0324
N7_anion_anion_019	C1	-2.6908	5.1080	7.7988
N7_anion_anion_020	C1	-3.1082	2.6928	5.8010
N7_cation_cation_001	Cs	-16.8094	-7.0687	9.7407
N7_cation_cation_003	Cs	-18.3210	-7.6492	10.6718
N7_cation_cation_005	Cs	-17.9319	-10.5556	7.3763
N7_cation_cation_006	C1	-19.0163	-9.2300	9.7863
N8_neutral_neutral_002	C1	-11.0090	-0.5306	10.4784
N8_neutral_neutral_001	D2h	-11.2755	-0.0572	11.2183
N8_neutral_neutral_005	C2	-9.6080	-0.0360	9.5720
N8_neutral_neutral_003	C1	-9.8865	0.0206	9.9071
N8_neutral_neutral_008	C1	-12.1214	-1.0493	11.0721
N8_neutral_neutral_004	C1	-10.3083	-0.1899	10.1184
N8_neutral_neutral_009	D3h	-11.7827	-1.2276	10.5551
N8_neutral_neutral_010	Cs	-9.8734	-1.8161	8.0573
N8_neutral_neutral_007	C2v	-11.0850	-1.6728	9.4122
N8_neutral_neutral_012	Cs	-11.5405	-1.1895	10.3510
N8_neutral_neutral_013	C2	-10.3508	-1.5137	8.8371
N8_neutral_neutral_019	D2d	-12.0840	-1.8861	10.1979
N8_neutral_neutral_020	C2h	-11.2639	-0.9453	10.3186
N8_neutral_neutral_016	Cs	-11.5952	-0.6090	10.9862
N8_neutral_neutral_014	C1	-9.5640	-1.3244	8.2396
N8_neutral_neutral_021	C2	-11.1279	-0.8641	10.2638
N8_neutral_neutral_022	C1	-10.6047	-2.0242	8.5805
N8_neutral_neutral_024	D2h	-9.2373	-1.2856	7.9517
N8_neutral_neutral_023	D2h	-11.1234	-0.1510	10.9724
N8_neutral_neutral_025	Oh	-10.9248	-0.3384	10.5864
N9_anion_anion_001	C1	-3.3287	4.4403	7.7690
N9_anion_anion_006	Cs	-4.5780	4.6285	9.2065
N9_anion_anion_003	C1	-3.5542	4.7732	8.3274
N9_anion_anion_008	C1	-4.1793	4.1556	8.3349
N9_anion_anion_004	C1	-3.6804	4.9283	8.6087
N9_anion_anion_011	Cs	-5.1788	4.2483	9.4271
N9_anion_anion_005	C2	-4.3825	5.0316	9.4141
N9_anion_anion_010	C1	-3.8293	4.7130	8.5423
N9_anion_anion_012	C1	-3.8357	3.8009	7.6366
N9_anion_anion_013	C1	-3.1500	4.3463	7.4963
N9_anion_anion_018	C1	-3.7990	4.5277	8.3267
N9_anion_anion_019	C1	-2.9452	4.5838	7.5290
N9_anion_anion_016	C2v	-2.8218	4.4852	7.3070
N9_cation_cation_006	C2v	-18.0356	-7.4402	10.5954
N9_cation_cation_008	C1	-16.9933	-7.0034	9.9899
N9_cation_cation_009	C1	-17.0050	-8.3209	8.6841
N9_cation_cation_012	C2v	-17.6775	-7.8159	9.8616
N9_cation_cation_011	Cs	-18.3662	-8.9103	9.4559
N10_neutral_neutral_001	C2	-12.9198	-0.5398	12.3800
N10_neutral_neutral_002	C1	-11.7475	-1.3053	10.4422
N10_neutral_neutral_003	C1	-9.9826	-1.1691	8.8135
N10_neutral_neutral_006	Cs	-11.2573	-0.6247	10.6326
N10_neutral_neutral_004	C1	-9.7070	-1.4443	8.2627
N10_neutral_neutral_007	Cs	-10.5916	-1.8512	8.7404
N10_neutral_neutral_005	C1	-10.3189	-1.6633	8.6556
N10_neutral_neutral_011	C1	-10.2546	-1.8095	8.4451
N10_neutral_neutral_010	C1	-11.1858	-0.6563	10.5295
N10_neutral_neutral_015	C2	-10.7038	-1.6499	9.0539

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N10_neutral_neutral_016	C1	-10.7642	-1.5672	9.1970
N10_neutral_neutral_014	C1	-10.1760	-3.1308	7.0452
N10_neutral_neutral_017	C2v	-11.2487	-1.3285	9.9202
N10_neutral_neutral_021	C2v	-11.2774	-1.0998	10.1776
N10_neutral_neutral_013	C2v	-8.7777	-2.5200	6.2577
N10_neutral_neutral_018	C2	-11.1371	0.6545	11.7916
N10_neutral_neutral_019	C3v	-11.3917	0.3446	11.7363
N10_neutral_neutral_020	C2	-10.5955	-1.4418	9.1537
N10_neutral_neutral_022	D5h	-11.4974	-0.8667	10.6307
N11_anion_anion_001	C1	-4.6868	4.7575	9.4443
N11_anion_anion_002	C1	-4.6917	4.4152	9.1069
N11_anion_anion_006	C1	-4.2089	4.0149	8.2238
N11_anion_anion_005	C1	-4.0097	4.1648	8.1745
N11_anion_anion_004	C1	-4.1785	4.4585	8.6370
N11_anion_anion_007	Cs	-4.6404	4.6470	9.2874
N11_anion_anion_010	Cs	-3.3157	4.1356	7.4513
N11_anion_anion_008	C1	-3.9921	4.4067	8.3988
N11_anion_anion_003	C2	-4.8000	3.0674	7.8674
N11_anion_anion_013	C1	-2.7734	2.6705	5.4439
N11_anion_anion_017	C1	-4.0058	4.4202	8.4260
N11_cation_cation_002	C1	-15.7017	-6.1480	9.5537
N11_cation_cation_001	C2v	-16.4400	-6.4750	9.9650
N11_cation_cation_005	C2	-17.0074	-7.0585	9.9489
N11_cation_cation_004	C1	-16.4230	-7.7299	8.6931
N11_cation_cation_006	C1	-15.6724	-7.1994	8.4730
N11_cation_cation_007	C1	-16.4756	-7.3306	9.1450
N12_neutral_neutral_001	C1	-12.3196	-2.4745	9.8451
N12_neutral_neutral_003	C1	-11.9678	-1.5248	10.4430
N12_neutral_neutral_002	C1	-11.3338	-2.5898	8.7440
N12_neutral_neutral_004	Cs	-9.8374	-2.7778	7.0596
N12_neutral_neutral_006	C1	-10.3193	-1.8368	8.4825
N12_neutral_neutral_005	C1	-10.6336	-1.7561	8.8775
N12_neutral_neutral_007	D3d	-11.6168	-0.4953	11.1215
N12_neutral_neutral_008	C1	-10.9868	-0.5129	10.4739
N12_neutral_neutral_009	D6h	-10.5865	-0.9025	9.6840
N13_anion_anion_001	C1	-4.2327	3.0401	7.2728
N13_cation_cation_001	C1	-15.9024	-7.4824	8.4200
N13_cation_cation_004	C1	-17.1447	-7.0618	10.0829
N14_neutral_neutral_002	C1	-10.0340	-1.7495	8.2845
N14_neutral_neutral_001	C1	-9.4477	-2.1106	7.3371
N14_neutral_neutral_003	C2v	-10.3293	-2.3248	8.0045
N14_neutral_neutral_005	C2	-12.0686	0.6111	12.6797
N15_cation_cation_001	C1	-16.9221	-7.4624	9.4597
N15_cation_cation_002	C1	-17.0792	-5.2422	11.8370
N16_neutral_neutral_002	C1	-10.1771	-3.2447	6.9324
N16_neutral_neutral_001	C1	-9.7508	-3.0679	6.6829

6. Charges atomiques

Charges de Mulliken et Löwdin, natives à ORCA. Les charges de Bader (QTAIM) et de Manz (DDEC6) restent reportées (§9) – ni l'une ni l'autre n'est native à ORCA.

Tableau 9: Charges atomiques (Mulliken et Löwdin) de l'état fondamental confirmé de chaque espèce moléculaire (groupe formule, charge) – min/max sur les atomes N, classées par charge (neutre, cation, anion) puis par N croissant.

Nom	Charge	Mulliken min/max	Löwdin min/max
N4_neutral_neutral_001	0	-0.000 / 0.000	-0.000 / 0.000
N6_neutral_neutral_001	0	-0.552 / 1.088	-0.280 / 0.279
N8_neutral_neutral_002	0	-0.640 / 1.174	-0.450 / 0.341
N10_neutral_neutral_001	0	-0.282 / 0.818	-0.303 / 0.133
N12_neutral_neutral_001	0	-0.319 / 1.408	-0.380 / 0.154

(suite)

N14_neutral_neutral_002	0	-0.641 / 1.355	-0.508 / 0.371
N16_neutral_neutral_002	0	-0.584 / 1.489	-0.397 / 0.141
N5_cation_cation_003	+	-0.409 / 0.929	-0.069 / 0.465
N7_cation_cation_001	+	-0.296 / 1.030	-0.060 / 0.447
N9_cation_cation_006	+	-0.266 / 0.775	-0.184 / 0.263
N11_cation_cation_001	+	-0.656 / 1.068	-0.235 / 0.410
N12_cation_cation_001	+	-0.169 / 1.074	-0.269 / 0.207
N13_cation_cation_001	+	-0.293 / 0.835	-0.181 / 0.218
N15_cation_cation_001	+	-0.422 / 0.873	-0.258 / 0.419
N5_anion_anion_001	-	-0.200 / -0.200	-0.200 / -0.200
N7_anion_anion_002	-	-0.853 / 0.992	-0.389 / 0.103
N9_anion_anion_001	-	-0.818 / 1.443	-0.568 / 0.159
N11_anion_anion_006	-	-0.756 / 1.131	-0.375 / 0.197
N13_anion_anion_001	-	-0.979 / 1.392	-0.394 / 0.224

7. Distances et ordres de liaison N–N : corrélation avec ΔH_f

Pour les composés neutres, proportion de liaisons courtes (triple, double, ou délocalisées $\sim 1{,}5$) parmi les liaisons réelles, et ordre de Mayer moyen, en regard du ΔH_f GFN2-xTB. Tendance attendue : davantage de caractère multiple devrait corréler avec un ΔH_f plus élevé (structures plus contraintes, moins proches de 2 N_2) – à confirmer statistiquement une fois la campagne DFT complète (l'échantillon actuel, 45 structures neutres, montre la tendance qualitativement mais reste bruité).

Tableau 10: Composés **neutres** terminés : proportion de liaisons N–N courtes (triple/double/délocalisées) parmi les liaisons réelles (hors contacts non-liants), et ΔH_f GFN2-xTB. Seuils de classification en §9 de `harvest_results.py`.

Nom	liaisons	BO Mayer moy.	% courtes[1]	classes présentes	ΔH_f^{xtb}
N10_neutral_neutral_001	11	0.806	36%	delocalized 1.5, single	-134.2
N10_neutral_neutral_002	10	0.917	70%	delocalized 1.5, single, triple	-123.7
N10_neutral_neutral_003	9	1.071	67%	delocalized 1.5, single, triple	-110.7
N10_neutral_neutral_004	10	0.829	60%	delocalized 1.5, single, triple	-100.84
N10_neutral_neutral_005	10	0.937	50%	delocalized 1.5, double, single	-97.64
N10_neutral_neutral_006	12	0.792	25%	double, single	-95.64
N10_neutral_neutral_007	11	0.790	36%	delocalized 1.5, single	-80.55
N10_neutral_neutral_010	12	0.772	25%	delocalized 1.5, single	-45.58
N10_neutral_neutral_011	11	0.787	45%	delocalized 1.5, double, single	-41.25
N10_neutral_neutral_013	12	0.643	33%	delocalized 1.5, single	-34.65
N10_neutral_neutral_014	12	0.713	42%	delocalized 1.5, long/weak, single	-33.7
N10_neutral_neutral_015	12	0.768	25%	delocalized 1.5, single	-22.96
N10_neutral_neutral_016	12	0.760	25%	delocalized 1.5, double, single	-20.37
N10_neutral_neutral_017	13	0.733	15%	double, single	-3.74
N10_neutral_neutral_018	15	0.658	0%	single	8.82
N10_neutral_neutral_019	15	0.666	0%	single	13.77
N10_neutral_neutral_020	14	0.671	7%	delocalized 1.5, single	34.26
N10_neutral_neutral_021	13	0.781	15%	delocalized 1.5, single	40.4
N10_neutral_neutral_022	15	0.663	0%	single	49.74
N12_neutral_neutral_001	13	0.615	38%	delocalized 1.5, single	-147.92

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N12_neutral_neutral_002	13	0.700	62%	delocalized 1.5, single	-134.68
N12_neutral_neutral_003	13	0.801	69%	delocalized 1.5, double, single	-112.01
N12_neutral_neutral_004	13	0.695	38%	delocalized 1.5, single	-102.34
N12_neutral_neutral_005	14	0.760	29%	double, single	-79.39
N12_neutral_neutral_006	14	0.722	29%	delocalized 1.5, double, single	-71.56
N12_neutral_neutral_007	18	0.645	0%	single	-46.96
N12_neutral_neutral_008	15	0.787	20%	delocalized 1.5, single	-14.76
N12_neutral_neutral_009	18	0.605	0%	single	94.51
N14_neutral_neutral_001	16	0.599	50%	delocalized 1.5, single	-140.21
N14_neutral_neutral_002	16	0.777	31%	delocalized 1.5, double, single, triple	-99.45
N14_neutral_neutral_003	17	0.702	24%	delocalized 1.5, double, single	-62.32
N14_neutral_neutral_005	21	0.681	0%	long/weak, single	-6.87
N16_neutral_neutral_001	18	0.478	44%	delocalized 1.5, single	-188.3
N16_neutral_neutral_002	17	0.665	53%	delocalized 1.5, single	-151.6
N4_neutral_neutral_001	6	0.796	0%	single	27.32
N4_neutral_neutral_002	4	1.118	50%	delocalized 1.5, double, single	53.79
N4_neutral_neutral_003	5	0.841	0%	long/weak, single	58.33
N4_neutral_neutral_004	4	1.056	50%	delocalized 1.5, single	68.96
N6_neutral_neutral_001	5	1.248	80%	delocalized 1.5, single, triple	-63.71
N6_neutral_neutral_002	6	0.835	100%	delocalized 1.5	-39.62
N6_neutral_neutral_003	6	1.112	50%	delocalized 1.5, double, single, triple	-8.71
N6_neutral_neutral_005	6	1.071	67%	delocalized 1.5, long/weak, triple	0.91
N6_neutral_neutral_006	6	1.133	67%	delocalized 1.5, long/weak, single, triple	15.09
N6_neutral_neutral_007	7	0.867	29%	delocalized 1.5, single	19.92
N6_neutral_neutral_008	8	0.790	12%	double, single	20.81
N6_neutral_neutral_009	7	0.979	29%	double, single	47.09
N6_neutral_neutral_010	6	0.935	33%	single, triple	50.62
N6_neutral_neutral_012	9	0.729	0%	single	72.95
N8_neutral_neutral_001	9	0.698	56%	delocalized 1.5, single	-113.38
N8_neutral_neutral_002	8	0.891	75%	delocalized 1.5, single, triple	-103.47
N8_neutral_neutral_003	7	1.024	71%	delocalized 1.5, single, triple	-93.12
N8_neutral_neutral_004	7	1.164	71%	delocalized 1.5, single, triple	-85.29
N8_neutral_neutral_005	7	1.158	71%	delocalized 1.5, single, triple	-83.97
N8_neutral_neutral_007	9	0.742	44%	delocalized 1.5, single	-47.0
N8_neutral_neutral_008	9	0.802	56%	delocalized 1.5, double, single	-46.16
N8_neutral_neutral_009	9	0.806	33%	double, single	-39.68
N8_neutral_neutral_010	8	0.910	50%	delocalized 1.5, single, triple	-29.32
N8_neutral_neutral_012	10	0.765	20%	double, single	-19.41
N8_neutral_neutral_013	9	0.831	67%	delocalized 1.5, long/weak, single	-10.48
N8_neutral_neutral_014	8	1.061	62%	delocalized 1.5, long/weak, single, triple	-0.15

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N8_neutral_neutral_016	11	0.743	9%	delocalized 1.5, single	19.4
N8_neutral_neutral_019	10	0.828	20%	double, single	36.92
N8_neutral_neutral_020	10	0.808	20%	delocalized 1.5, single	39.55
N8_neutral_neutral_021	10	0.791	20%	delocalized 1.5, single	42.28
N8_neutral_neutral_022	10	0.808	30%	delocalized 1.5, long/weak, single	48.47
N8_neutral_neutral_023	12	0.644	0%	single	51.3
N8_neutral_neutral_024	11	0.618	18%	delocalized 1.5, long/weak, single	69.67
N8_neutral_neutral_025	12	0.701	0%	single	92.32

Note : ΔH_f^{xtb} (composés **neutres** uniquement, cette table) est calculé pour la réaction $(n/2)\,\mathrm{N_2} \rightarrow \mathrm{N}_n$, soit $\Delta H_f = E(\mathrm{N}_n) - (n/2)\,E(\mathrm{N_2})$, référence $\Delta H_f(\mathrm{N_2}) = 0$. C'est une définition **différente** de celle utilisée pour les cations/anions (Tableaux 3–4, §2) : $\mathrm{N_5^-}(\mathrm{D}_{5h}) + ((n-5)/2)\,\mathrm{N_2} \rightarrow \mathrm{N}_n^-$ et $\mathrm{N_5^+}(\mathrm{C}_{2v}) + ((n-5)/2)\,\mathrm{N_2} \rightarrow \mathrm{N}_n^+$, ancrées sur les ΔH_f littérature de $\mathrm{N_5^-}/\mathrm{N_5^+}$ (valeurs §2) plutôt que sur N_2 seul – les deux définitions ne sont pas directement comparables entre elles.

8. Confrontation à la littérature et aux bases externes

Deux ensembles distincts sont comparés ici. **Nos 193 candidats** sont le jeu DFT WB97X-D4 de ce rapport : structures issues du criblage GFN2-xTB de `polyN-pipeline` (§1.1, Tableaux 2–4). **337 structures connues** désigne un ensemble séparé : le fichier `biblio_polyN/comparison_table_3sources.csv` de `polyN-pipeline`, qui catalogue des structures tirées d'articles publiés et des bases externes `N_csp` et `polyN_study`, toutes relaxées indépendamment au même niveau GFN2-xTB pour permettre la comparaison. Le croisement par (N, charge) entre nos 193 candidats et ces 337 références donne trois résultats complémentaires :

- **Couverture** : 12 des 19 groupes (89 des 193 structures, 46 %) n'ont **aucune** référence cataloguée à cette formule/charge.
- **Topologie** : pour les 104 candidats des 7 groupes comparables, une empreinte de graphe de liaisons (séquence de degrés + tailles de cycles + nombre de liaisons, seuil 1,90 Å, NetworkX) montre que 76 (73 %) ne correspondent à aucune connectivité cataloguée. Au total, **165 des 193 candidats (86 %)** sont donc distincts de tout ce qui existe dans le tableau à 3 sources.
- **Stabilité** : 3 des 7 groupes comparables voient leur meilleur candidat local dépasser en stabilité le meilleur catalogué – N_9^- ($-0{,}166$ eV/atome), N_{10} ($-0{,}066$, malgré 60 références disponibles), N_{12} ($-0{,}246$). N_6 et N_8 retombent exactement sur le minimum déjà connu. N_4 est la seule perte nette ($+0{,}302$ eV/atome). N_9^+ : le seul candidat terminé à cette composition (`N9_cation_cation_001`) est fragmenté en trois espèces séparées (2+2+5 atomes, Tableau S1) – aucune comparaison de stabilité n'est possible pour ce groupe tant qu'un candidat non-fragmenté n'a pas terminé.

Détail complet et graphique interactif : voir l'artefact *Uncharted Compositions*.

[1] Triple, double ou délocalisée (ordre $\sim 1{,}5$), à l'exclusion des liaisons simples et des contacts non-liants.

9. Limites et prochaines étapes

- La comparaison de stabilité de la section précédente reste au niveau **GFN2-xTB** – la confirmation définitive viendra des énergies WB97X-D4 elles-mêmes, une fois les 35 candidats non convergés repris avec davantage de cycles d'optimisation et les 38 candidats à fréquence imaginaire re-optimisés en vrais minima (§4).
- Charges de Bader (QTAIM) et de Manz (DDEC6) reportées à une étape ultérieure – nécessitent respectivement le code de Henkelman et Chargemol sur un cube de densité via `orca_plot`, non installés sur le cluster à ce jour.
- Tous les fichiers ORCA bruts (`.gbw`, `.densities`, `.hess`) sont conservés intacts. Une seconde campagne DLPNO-CCSD(T)-F12 (cc-pVTZ-F12, 49 structures représentatives – les 3 isomères DFT les plus stables de chacun des 19 groupes (N, charge) confirmés, plus N_2, NH_4^+ et NH_2^-) est en cours pour établir des statistiques de temps de calcul (humain/CPU) et le nombre de cœurs optimal par taille, avant d'étendre ce niveau à l'ensemble des candidats. Le module `refinement/` du dépôt compagnon `polyN` implémente déjà un enchaînement de méthodes config-driven avec seuil de taille (`max_n_atoms`) adapté à cet usage, à migrer vers plus tard.
- Seuils de classification des liaisons N–N (§6) : triple $< 1{,}15$ Å, double $< 1{,}22$, délocalisée $< 1{,}32$, simple $< 1{,}55$, au-delà non-liant – purement géométriques, à recouper avec l'ordre de Mayer (fourni en regard dans `results_bonds.csv`) plutôt qu'à prendre isolément.

A. Supporting Information

Matériel supplémentaire : tableaux (Tableau S1–S3) et figures (Figure S1–S5) numérotés à la suite, séparément des tableaux et sections du corps principal.

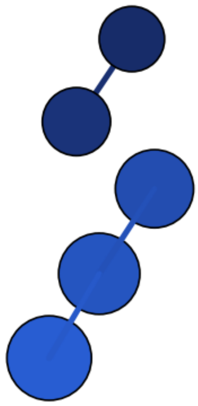
A.1 Structures fragmentées

Tableau S1. Structures fragmentées : le candidat se sépare en plusieurs espèces moléculaires distinctes (aucune paire N–N restante sous 1,7 Å entre les morceaux), pas un cluster N_x lié unique. N'apparaissent pas dans les Tableaux 2–4 (§2) ni dans la comparaison de stabilité DFT/xTB (§3). La colonne *déjà fragmenté à xTB* indique si la géométrie de départ (avant tout calcul DFT) était elle-même déjà séparée en plusieurs morceaux – auquel cas la fragmentation est antérieure à cette campagne DFT (artefact du criblage GFN2-xTB en amont), et non un résultat de l'optimisation DFT. Représentations avant/après en Figure S1.

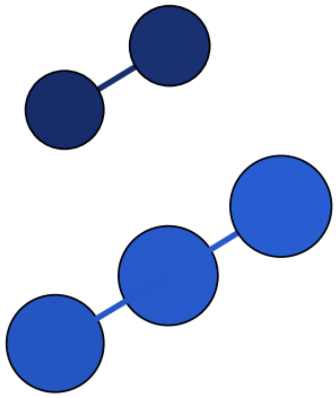
Nom	N	Charge	Fragments (tailles)	Niveau	Déjà frag. à xTB	Réf.
N5_anion_anion_002	5	-	2+3	DFT	✓ (2+3)	our work
N5_anion_anion_003	5	-	2+3	DFT	✓ (2+3)	our work
N5_anion_anion_005	5	-	2+3	DFT	✓ (2+3)	our work
N5_anion_anion_007	5	-	2+3	DFT	non (DFT seul)	our work
N5_anion_anion_009	5	-	2+3	DFT	✓ (2+3)	our work
N5_anion_anion_004	5	-	2+3	DFT	non (DFT seul)	our work
N5_anion_anion_008	5	-	2+3	DFT	non (DFT seul)	our work
N5_cation_cation_007	5	+	2+3	DFT	non (DFT seul)	our work
N6_neutral_neutral_004	6	0	2+2+2	DFT	non (DFT seul)	our work
N7_anion_anion_006	7	-	2+5	xtb	✓ (2+5)	our work
N8_neutral_neutral_006	8	0	3+5	DFT	non (DFT seul)	our work
N8_neutral_neutral_018	8	0	2+2+2+2	DFT	non (DFT seul)	our work
N9_anion_anion_007	9	-	3+6	xtb	✓ (3+6)	our work
N9_anion_anion_015	9	-	2+7	DFT	non (DFT seul)	our work
N9_anion_anion_009	9	-	2+7	xtb	✓ (2+7)	our work
N9_anion_anion_020	9	-	2+7	xtb	✓ (2+7)	our work
N9_cation_cation_001	9	+	2+2+5	DFT	non (DFT seul)	our work
N10_neutral_neutral_008	10	0	5+5	xtb	✓ (5+5)	our work
N10_neutral_neutral_009	10	0	3+7	xtb	✓ (3+7)	our work
N11_anion_anion_018	11	-	3+8	DFT	non (DFT seul)	our work
N11_anion_anion_011	11	-	3+3+5	DFT	✓ (3+3+5)	our work
N11_cation_cation_010	11	+	3+8	DFT	non (DFT seul)	our work
N11_cation_cation_009	11	+	3+8	DFT	non (DFT seul)	our work
N14_neutral_neutral_004	14	0	2+2+2+3+5	DFT	non (DFT seul)	our work

Figure S1 – structures fragmentées (avant / après)

Pour chaque candidat fragmenté (Tableau S1) : à gauche la géométrie initiale telle que produite par le criblage GFN2-xTB, à droite le résultat de l'optimisation DFT WB97X-D4. La légende sous chaque paire précise si la géométrie initiale était déjà fragmentée à xTB (artefact du criblage en amont) ou encore intacte (fragmentation survenue pendant l'optimisation DFT elle-même).

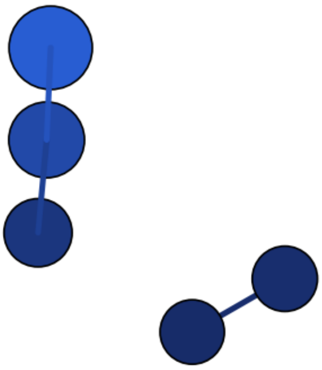


initiale (xTB, déjà fragmentée, 2+3)

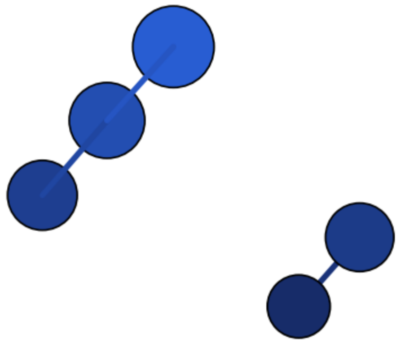


DFT (fragmentée, 2+3)

`N5_anion_anion_002` – fragments de taille 2+3

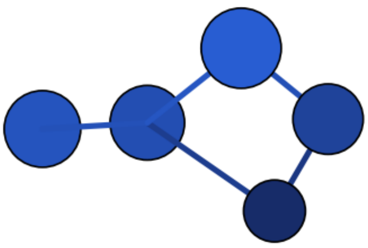


initiale (xTB, déjà fragmentée, 2+3)

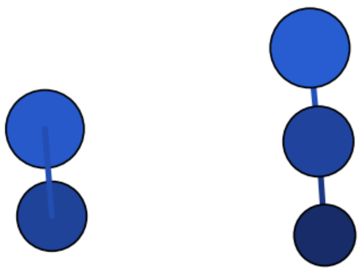


DFT (fragmentée, 2+3)

`N5_anion_anion_003` – fragments de taille 2+3

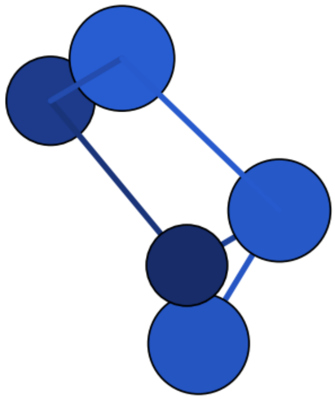


initiale (xTB, intacte, N_5^-)

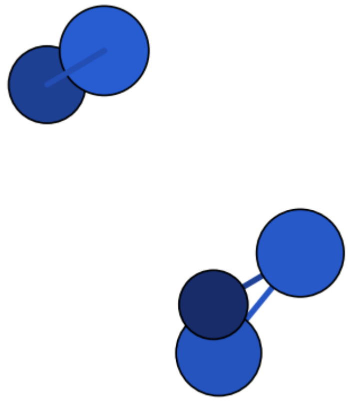


DFT (fragmentée, 2+3)

`N5_anion_anion_004` – fragments de taille 2+3

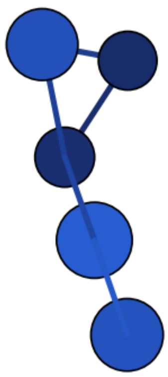


initiale (xTB, déjà fragmentée, 2+3)

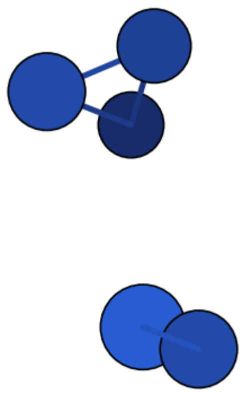


DFT (fragmentée, 2+3)

`N5_anion_anion_005` – fragments de taille 2+3

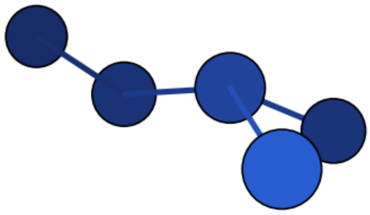


initiale (xTB, intacte, N_5^-)

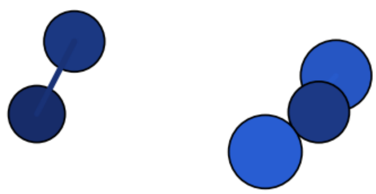


DFT (fragmentée, 2+3)

`N5_anion_anion_007` – fragments de taille 2+3

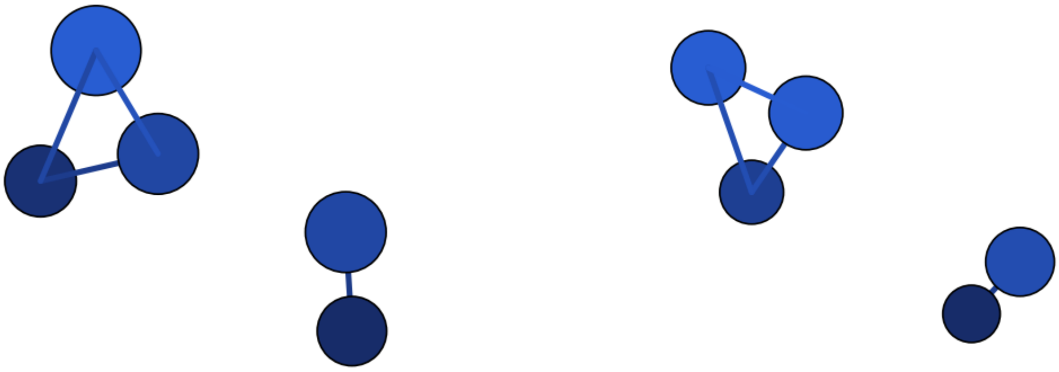


initiale (xTB, intacte, N_5^-)



DFT (fragmentée, 2+3)

`N5_anion_anion_008` – fragments de taille 2+3



initiale (xTB, déjà fragmentée, 2+3) DFT (fragmentée, 2+3)

`N5_anion_anion_009` – fragments de taille 2+3

initiale (xTB, intacte, N_5^+) DFT (fragmentée, 2+3)

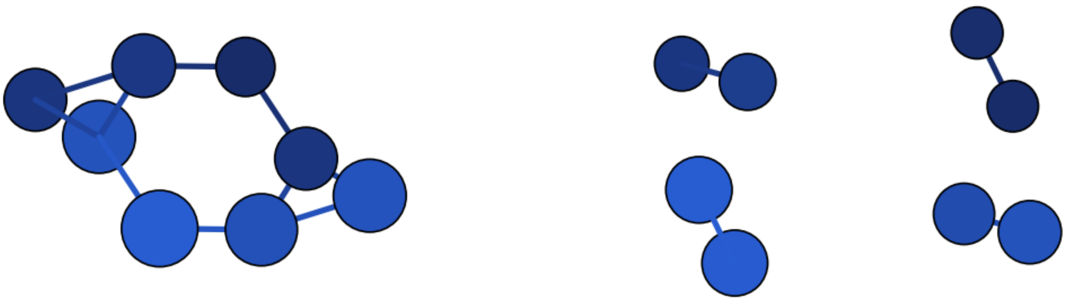
`N5_cation_cation_007` – fragments de taille 2+3

initiale (xTB, intacte, N_6^0) DFT (fragmentée, 2+2+2)

`N6_neutral_neutral_004` – fragments de taille 2+2+2

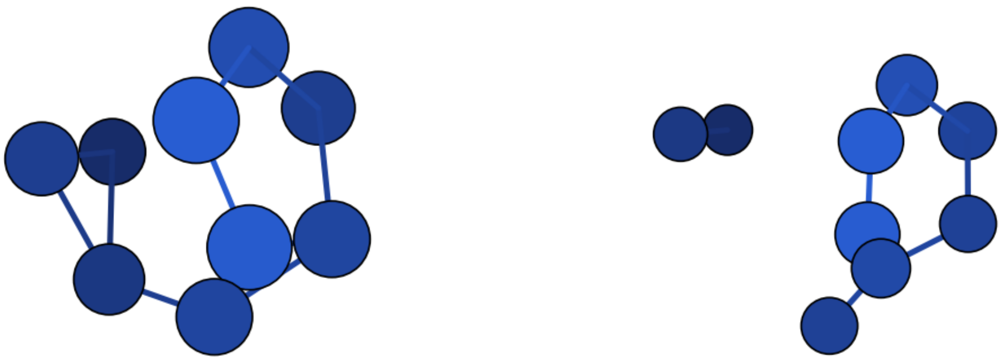
initiale (xTB, intacte, N_8^0) DFT (fragmentée, 3+5)

`N8_neutral_neutral_006` – fragments de taille 3+5



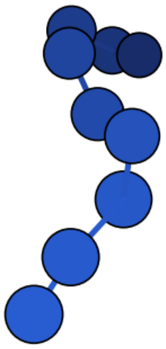
initiale (xTB, intacte, N_8^0) DFT (fragmentée, 2+2+2+2)

`N8_neutral_neutral_018` – fragments de taille 2+2+2+2

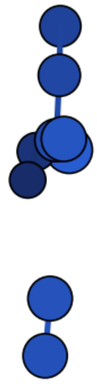


initiale (xTB, intacte, N_9^-) DFT (fragmentée, 2+7)

`N9_anion_anion_015` – fragments de taille 2+7

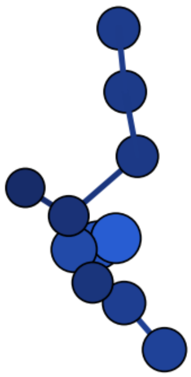


initiale (xTB, intacte, N_9^+)

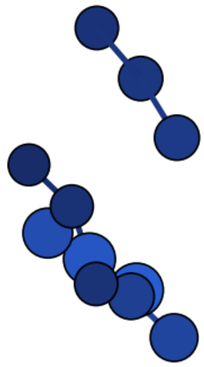


DFT (fragmentée, 2+2+5)

N9_cation_cation_001 – fragments de taille 2+2+5

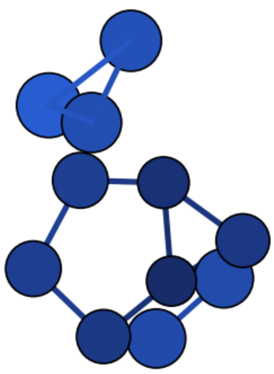


initiale (xTB, déjà fragmentée, 3+3+5)

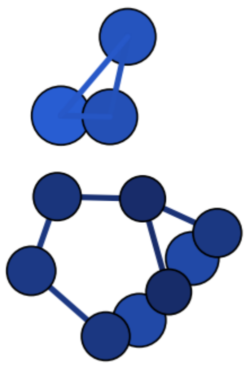


DFT (fragmentée, 3+3+5)

N11_anion_anion_011 – fragments de taille 3+3+5

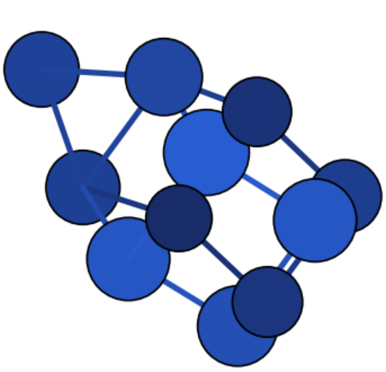


initiale (xTB, intacte, $N_{1}1^{-}$)

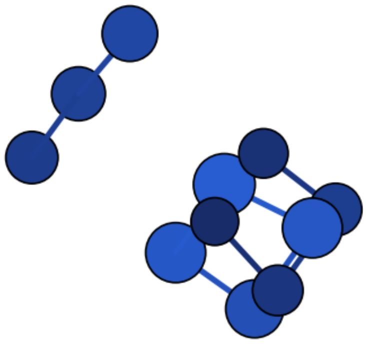


DFT (fragmentée, 3+8)

N11_anion_anion_018 – fragments de taille 3+8

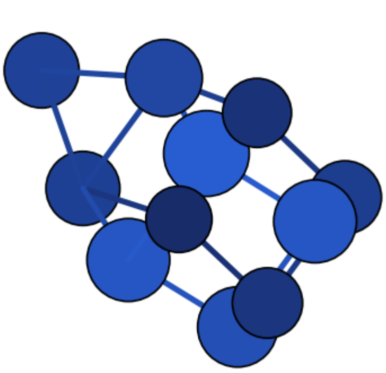


initiale (xTB, intacte, N$_{1}1^{+}$)

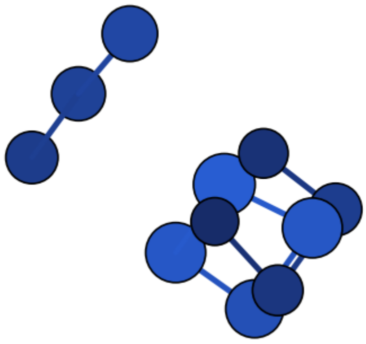


DFT (fragmentée, 3+8)

`N11_cation_cation_009` – fragments de taille 3+8

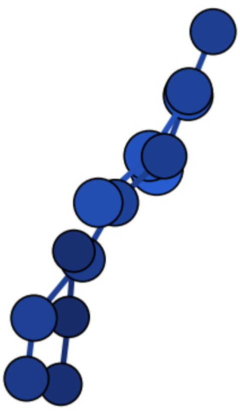


initiale (xTB, intacte, N$_{1}1^{+}$)

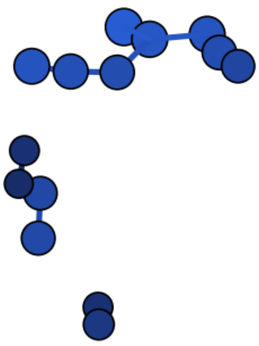


DFT (fragmentée, 3+8)

`N11_cation_cation_010` – fragments de taille 3+8



initiale (xTB, intacte, N$_{1}4^{0}$)



DFT (fragmentée, 2+2+2+3+5)

`N14_neutral_neutral_004` – fragments de taille 2+2+2+3+5

Légende générale, Figure S1 : sphères bleues reliées par des bâtonnets = atomes N et liaisons N–N (modèle boules-bâtonnets, rendu depuis les coordonnées cartésiennes .xyz) ; paires groupées par candidat, géométrie initiale GFN2-xTB à gauche et géométrie finale DFT WB97X-D4 à droite pour chacun des 18 candidats fragmentés dont le calcul DFT est terminé (sur les 24 recensés au Tableau S1 ; les autres n'ont pas encore de géométrie DFT à montrer).

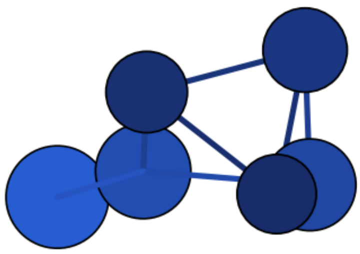
A.2 Structures non convergées

Tableau S2. Structures dont l'optimisation géométrique DFT WB97X-D4/aug-cc-pVTZ n'a **pas convergé** – l'optimiseur a atteint sa limite de 50 cycles (`The optimization did not converge but reached the maximum`) sans trouver de point stationnaire. Énergie non fiable, structures exclues de tous les tableaux d'analyse (§2–7) et de la comparaison de stabilité (§8) tant qu'elles n'ont pas été reprises. **28 des 35 n'ont en réalité probablement jamais de minimum lié** : la colonne *Frag. dernier cycle* montre qu'elles sont déjà séparées en plusieurs composantes connexes (seuil 1,70 Å) à la dernière géométrie tentée – l'optimiseur suit un gradient de dissociation qui ne s'annule jamais en un nombre fini de cycles, plutôt qu'un problème numérique de convergence ; les relancer avec `%geom MaxIter` plus grand ne changera probablement rien. Seules les 7 restantes (composé lié unique au dernier cycle) sont des candidates crédibles à une reprise avec davantage de cycles. Représentations avant/dernier cycle en Figure S2.

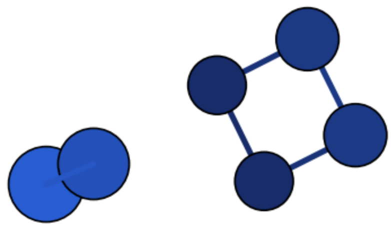
Nom	N	Charge	Cycles	Frag. dernier cycle	Réf.
`N6_neutral_neutral_011`	6	0	50	✓ (2+4)	our work
`N7_anion_anion_006`	7	-	50	✓ (2+2+3)	our work
`N7_anion_anion_015`	7	-	50	✓ (2+2+3)	our work
`N7_cation_cation_002`	7	+	50	✓ (2+5)	our work
`N7_cation_cation_004`	7	+	50	✓ (2+2+3)	our work
`N8_neutral_neutral_011`	8	0	50	non	our work
`N8_neutral_neutral_015`	8	0	50	non	our work
`N8_neutral_neutral_017`	8	0	50	✓ (2+6)	our work
`N9_anion_anion_002`	9	-	50	✓ (2+2+5)	our work
`N9_anion_anion_007`	9	-	50	✓ (3+6)	our work
`N9_anion_anion_009`	9	-	50	✓ (2+7)	our work
`N9_anion_anion_014`	9	-	50	✓ (2+7)	our work
`N9_anion_anion_017`	9	-	50	non	our work
`N9_anion_anion_020`	9	-	50	✓ (2+7)	our work
`N9_cation_cation_002`	9	+	50	✓ (2+2+5)	our work
`N9_cation_cation_003`	9	+	50	✓ (2+7)	our work
`N9_cation_cation_004`	9	+	50	✓ (2+2+5)	our work
`N9_cation_cation_005`	9	+	50	✓ (2+2+2+3)	our work
`N9_cation_cation_007`	9	+	50	✓ (2+2+5)	our work
`N9_cation_cation_010`	9	+	50	non	our work
`N10_neutral_neutral_008`	10	0	50	✓ (2+2+3+3)	our work
`N10_neutral_neutral_009`	10	0	50	✓ (2+8)	our work
`N10_neutral_neutral_012`	10	0	50	✓ (2+2+6)	our work
`N11_anion_anion_009`	11	-	50	✓ (2+2+7)	our work
`N11_anion_anion_012`	11	-	50	non	our work
`N11_anion_anion_014`	11	-	50	✓ (2+9)	our work
`N11_anion_anion_015`	11	-	50	non	our work
`N11_anion_anion_016`	11	-	50	✓ (2+2+7)	our work
`N11_cation_cation_003`	11	+	50	✓ (2+2+7)	our work
`N11_cation_cation_008`	11	+	50	✓ (2+2+2+5)	our work
`N13_anion_anion_002`	13	-	50	✓ (2+11)	our work
`N13_anion_anion_003`	13	-	50	non	our work
`N13_anion_anion_004`	13	-	50	✓ (2+11)	our work
`N13_cation_cation_002`	13	+	50	✓ (2+2+2+7)	our work
`N13_cation_cation_003`	13	+	50	✓ (2+2+9)	our work

Figure S2 – structures non convergées (avant / dernier cycle)

Pour chaque candidat non convergé (Tableau S2) : à gauche la géométrie initiale (criblage GFN2-xTB), à droite la dernière géométrie tentée par l'optimiseur DFT avant d'atteindre la limite de cycles – pas un minimum, juste le dernier point de la trajectoire d'optimisation.

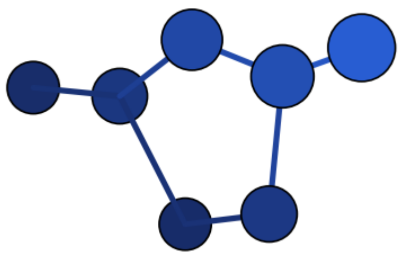


initiale (xTB, N_6^0)

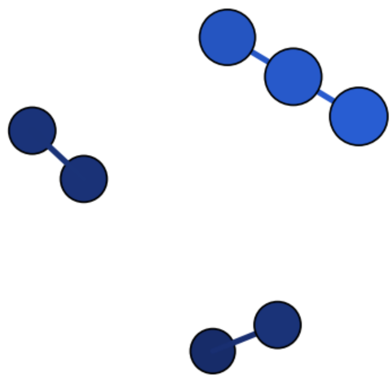


dernier cycle DFT (50 cycles)

`N6_neutral_neutral_011` – en cours de fragmentation, 2+4



initiale (xTB, N_7^-)



dernier cycle DFT (50 cycles)

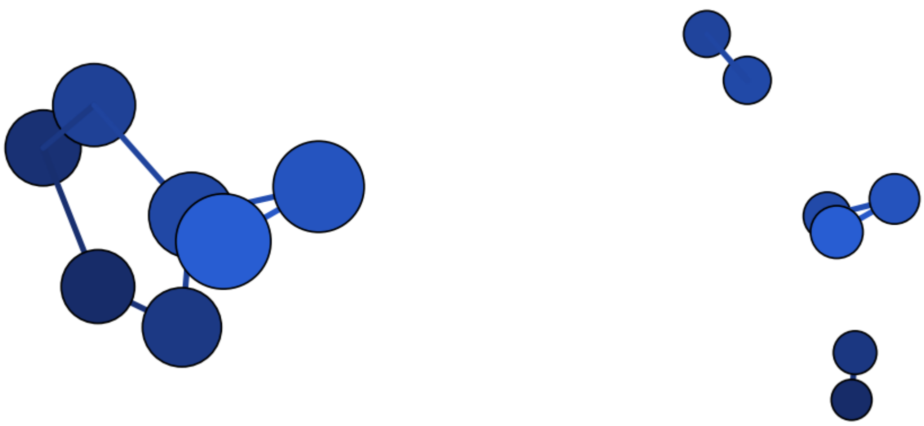
`N7_anion_anion_006` – en cours de fragmentation, 2+2+3

initiale (xTB, N_7^-) dernier cycle DFT (50 cycles)

`N7_anion_anion_015` – en cours de fragmentation, 2+2+3

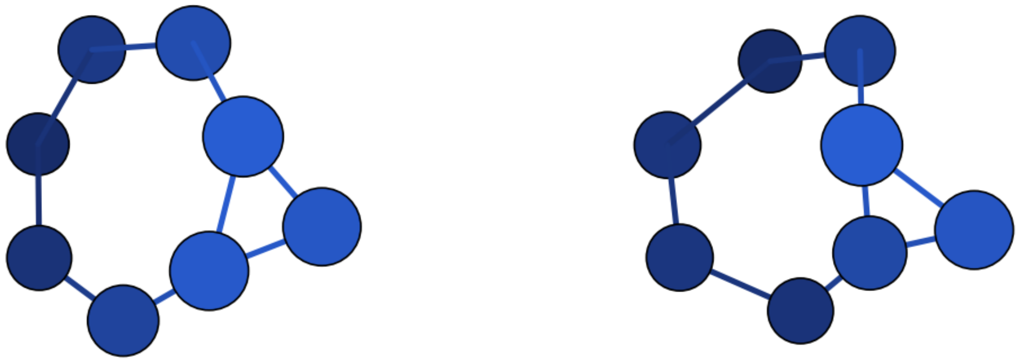
initiale (xTB, N_7^+) dernier cycle DFT (50 cycles)

`N7_cation_cation_002` – en cours de fragmentation, 2+5



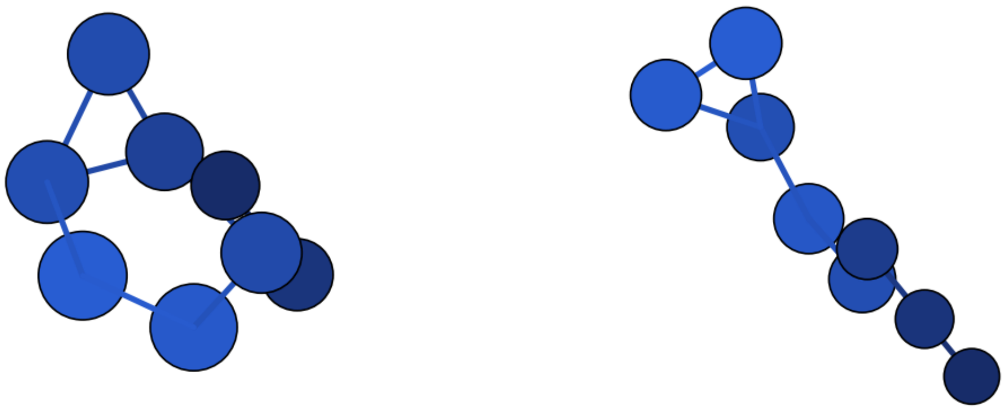
initiale (xTB, N_7^+) dernier cycle DFT (50 cycles)

`N7_cation_cation_004` – en cours de fragmentation, 2+2+3



initiale (xTB, N_8^0) dernier cycle DFT (50 cycles)

`N8_neutral_neutral_011` – cluster unique – non-convergence numérique

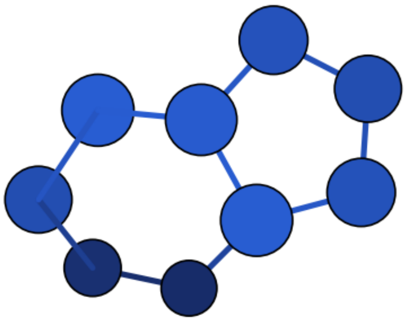


initiale (xTB, N_8^0) dernier cycle DFT (50 cycles)

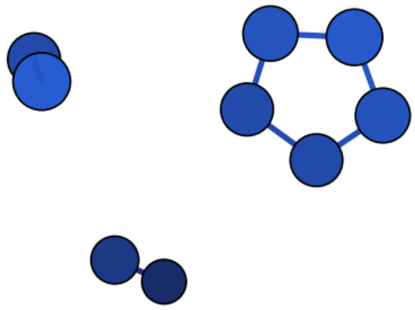
`N8_neutral_neutral_015` – cluster unique – non-convergence numérique

initiale (xTB, N_8^0) dernier cycle DFT (50 cycles)

`N8_neutral_neutral_017` – en cours de fragmentation, 2+6

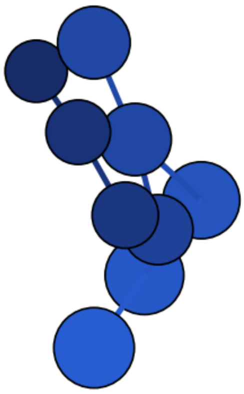


initiale (xTB, N_9^-)

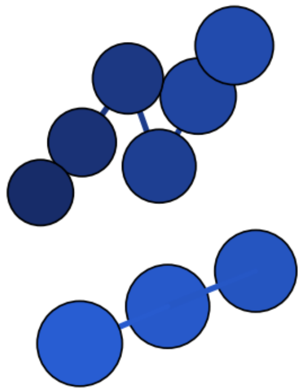


dernier cycle DFT (50 cycles)

`N9_anion_anion_002` – en cours de fragmentation, 2+2+5

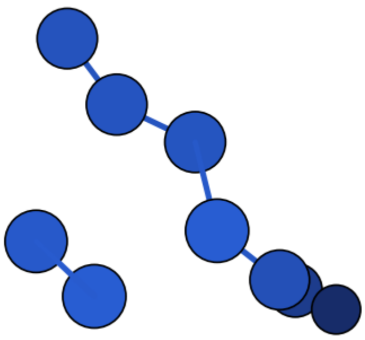


initiale (xTB, N_9^-)

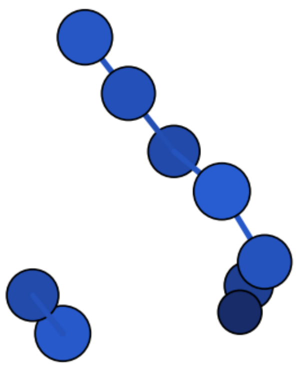


dernier cycle DFT (50 cycles)

`N9_anion_anion_007` – en cours de fragmentation, 3+6

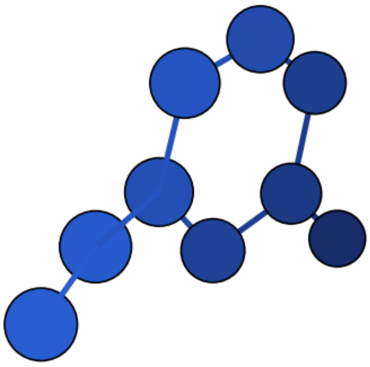


initiale (xTB, N_9^-)

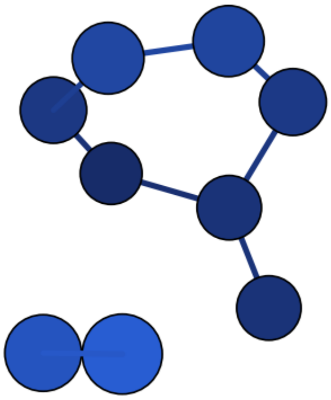


dernier cycle DFT (50 cycles)

`N9_anion_anion_009` – en cours de fragmentation, 2+7

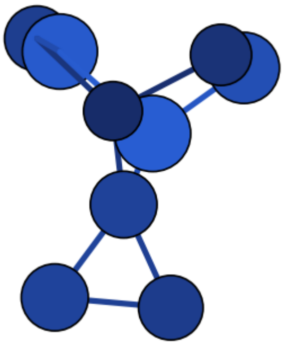


initiale (xTB, N_9^-)

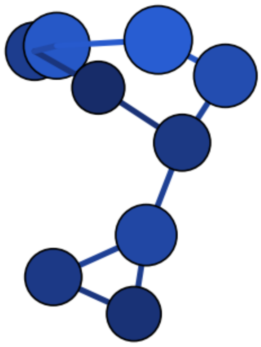


dernier cycle DFT (50 cycles)

`N9_anion_anion_014` – en cours de fragmentation, 2+7

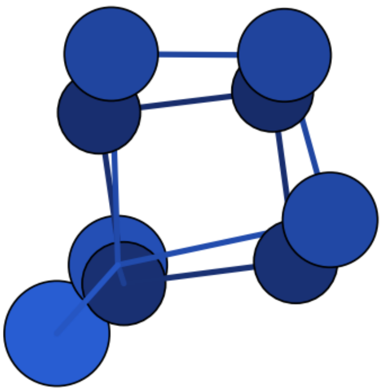


initiale (xTB, N_9^-)

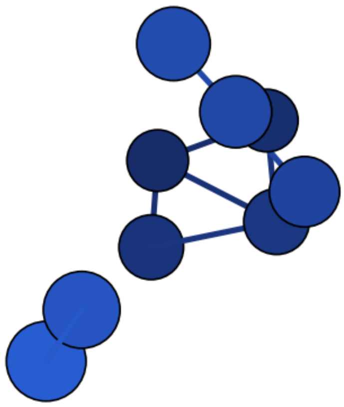


dernier cycle DFT (50 cycles)

`N9_anion_anion_017` – cluster unique – non-convergence numérique

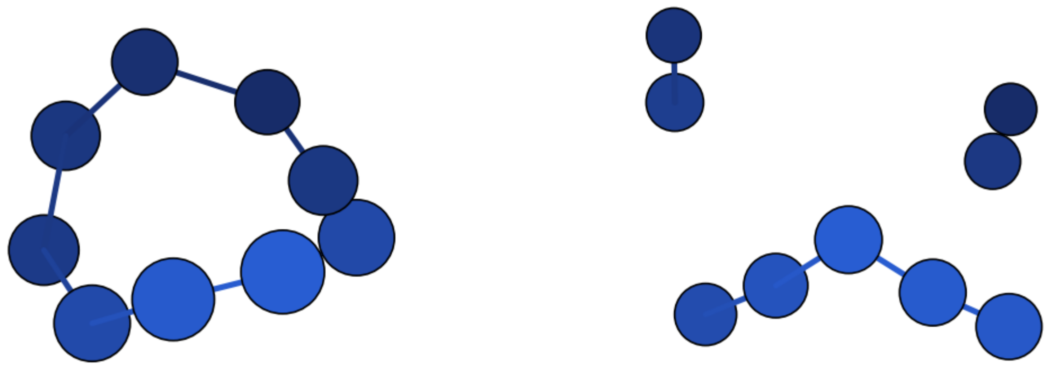


initiale (xTB, N_9^-)



dernier cycle DFT (50 cycles)

`N9_anion_anion_020` – en cours de fragmentation, 2+7



initiale (xTB, N_9^+) dernier cycle DFT (50 cycles)

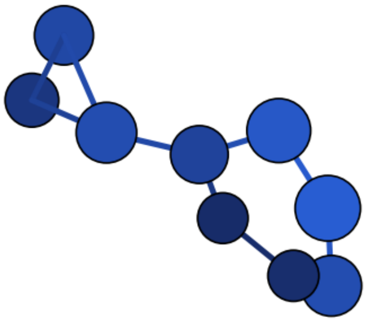
`N9_cation_cation_002` – en cours de fragmentation, 2+2+5

initiale (xTB, N_9^+) dernier cycle DFT (50 cycles)

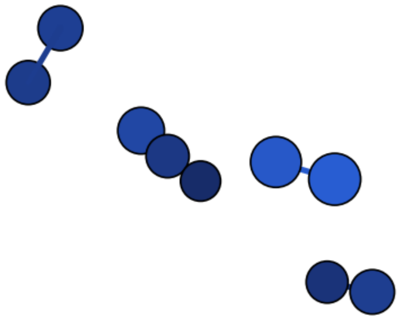
`N9_cation_cation_003` – en cours de fragmentation, 2+7

initiale (xTB, N_9^+) dernier cycle DFT (50 cycles)

`N9_cation_cation_004` – en cours de fragmentation, 2+2+5

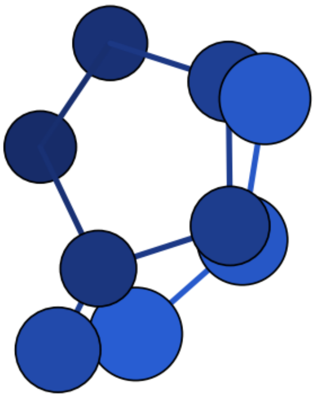


initiale (xTB, N_9^+)

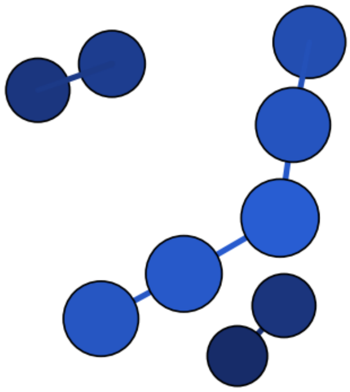


dernier cycle DFT (50 cycles)

N9_cation_cation_005 – en cours de fragmentation, 2+2+2+3

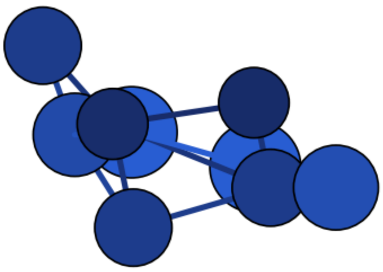


initiale (xTB, N_9^+)

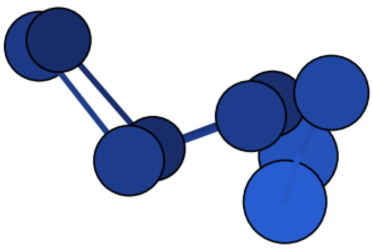


dernier cycle DFT (50 cycles)

N9_cation_cation_007 – en cours de fragmentation, 2+2+5

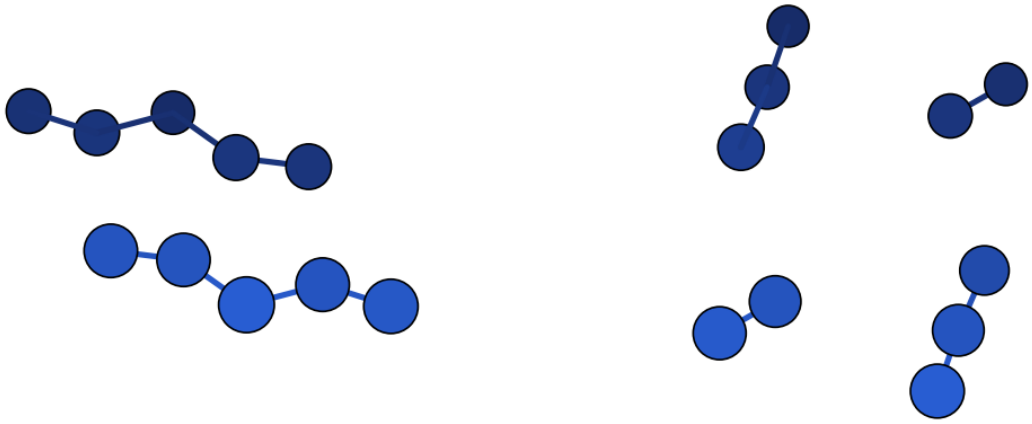


initiale (xTB, N_9^+)



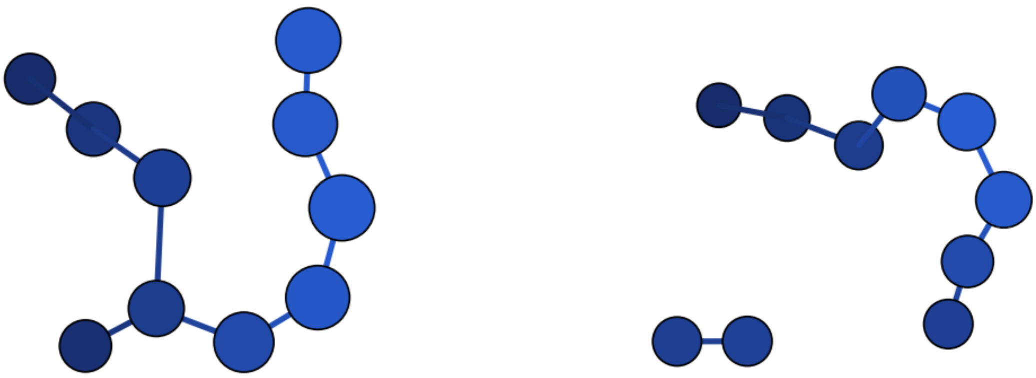
dernier cycle DFT (50 cycles)

N9_cation_cation_010 – cluster unique – non-convergence numérique



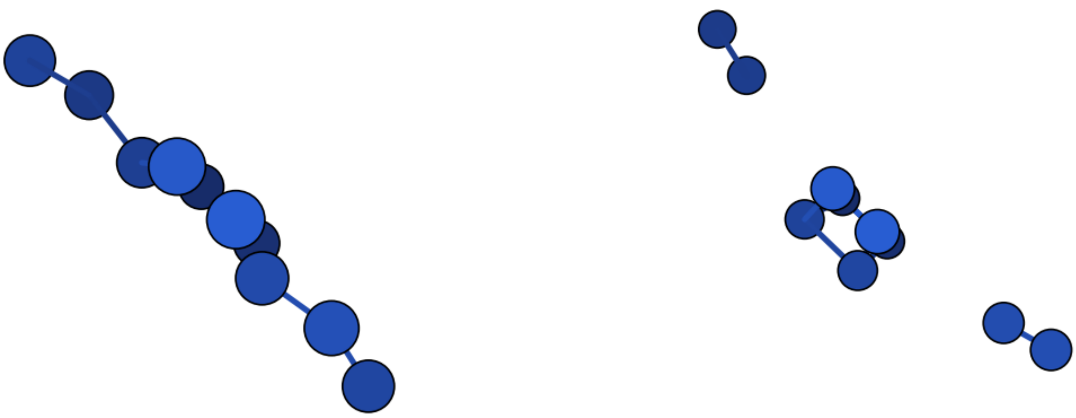
initiale (xTB, N$_{1}0^{0}$) dernier cycle DFT (50 cycles)

N10_neutral_neutral_008 – en cours de fragmentation, 2+2+3+3



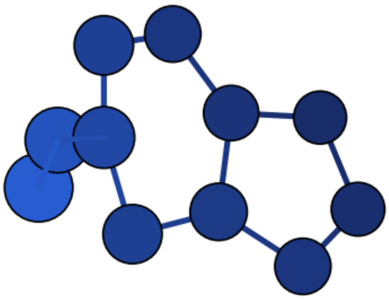
initiale (xTB, N$_{1}0^{0}$) dernier cycle DFT (50 cycles)

N10_neutral_neutral_009 – en cours de fragmentation, 2+8

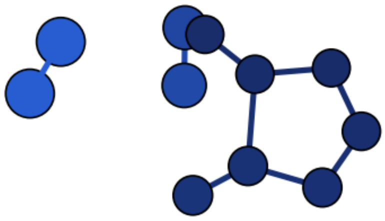


initiale (xTB, N$_{1}0^{0}$) dernier cycle DFT (50 cycles)

N10_neutral_neutral_012 – en cours de fragmentation, 2+2+6

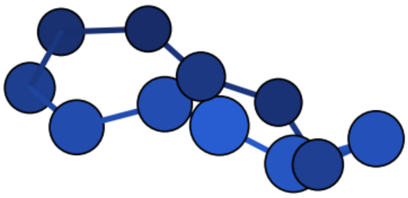


initiale (xTB, $N_{1}1^{-}$)

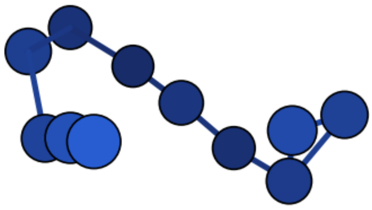


dernier cycle DFT (50 cycles)

N11_anion_anion_009 – en cours de fragmentation, 2+2+7

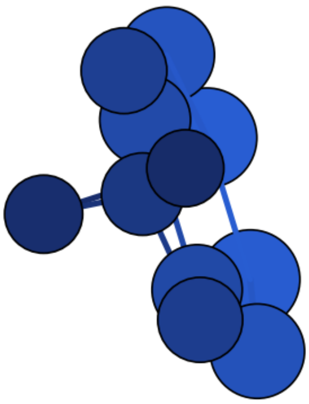


initiale (xTB, $N_{1}1^{-}$)

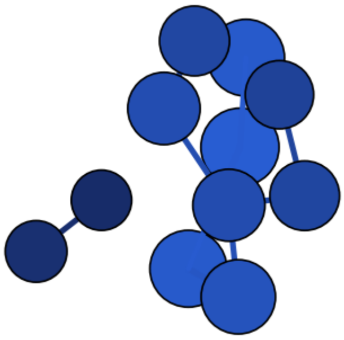


dernier cycle DFT (50 cycles)

N11_anion_anion_012 – cluster unique – non-convergence numérique

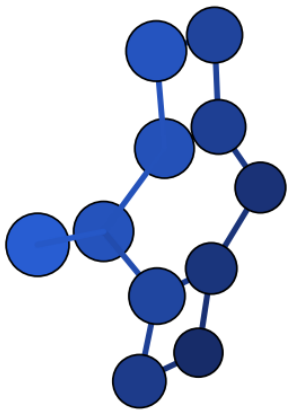


initiale (xTB, $N_{1}1^{-}$)

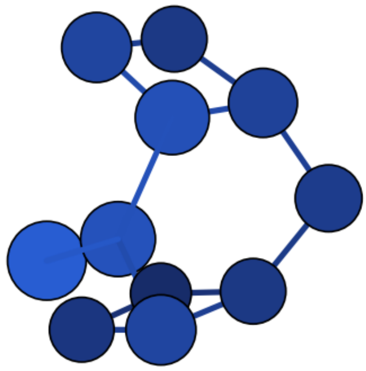


dernier cycle DFT (50 cycles)

N11_anion_anion_014 – en cours de fragmentation, 2+9

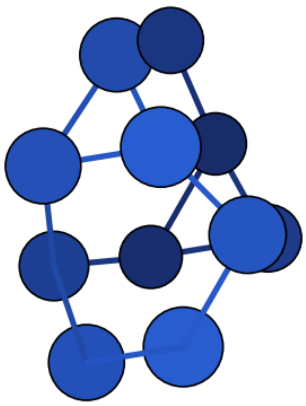


initiale (xTB, N_11^-)

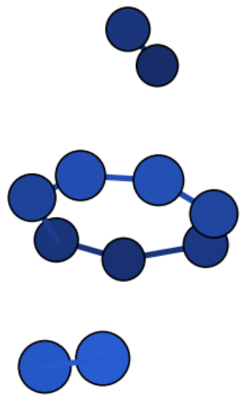


dernier cycle DFT (50 cycles)

`N11_anion_anion_015` – cluster unique – non-convergence numérique

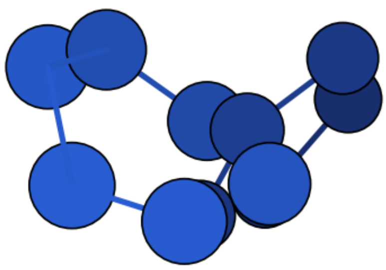


initiale (xTB, N_11^-)

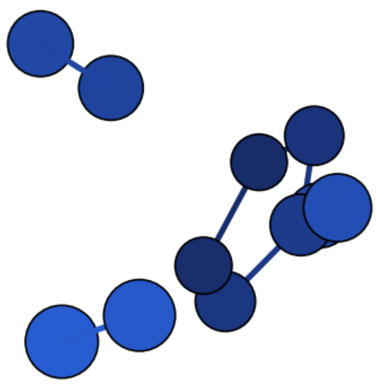


dernier cycle DFT (50 cycles)

`N11_anion_anion_016` – en cours de fragmentation, 2+2+7

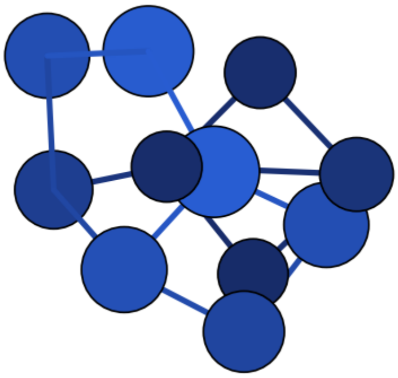


initiale (xTB, N_11^+)

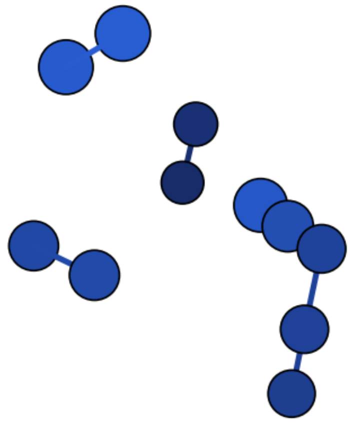


dernier cycle DFT (50 cycles)

`N11_cation_cation_003` – en cours de fragmentation, 2+2+7

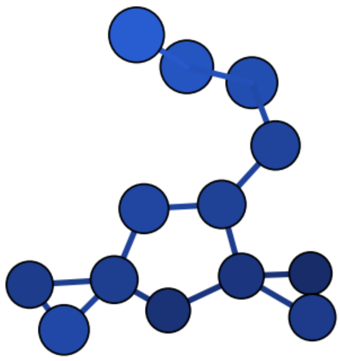


initiale (xTB, $N_1 1^+$)

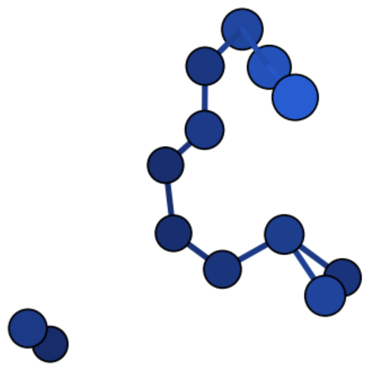


dernier cycle DFT (50 cycles)

`N11_cation_cation_008` – en cours de fragmentation, 2+2+2+5

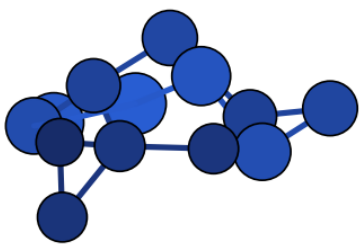


initiale (xTB, $N_1 3^-$)

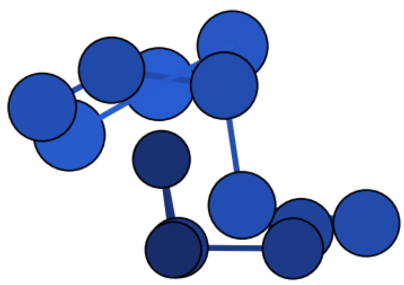


dernier cycle DFT (50 cycles)

`N13_anion_anion_002` – en cours de fragmentation, 2+11

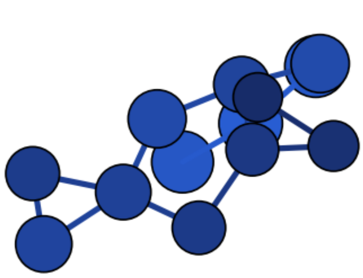


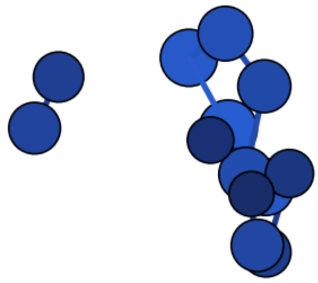
initiale (xTB, $N_1 3^-$)



dernier cycle DFT (50 cycles)

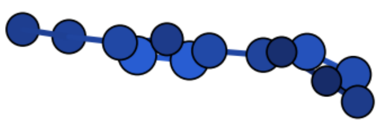
`N13_anion_anion_003` – cluster unique – non-convergence numérique

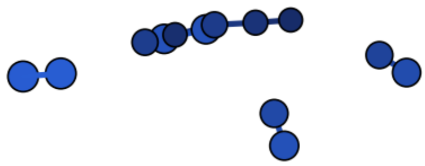




initiale (xTB, N_13^{-}) dernier cycle DFT (50 cycles)

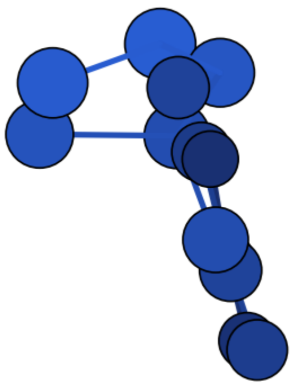
`N13_anion_anion_004` – en cours de fragmentation, 2+11

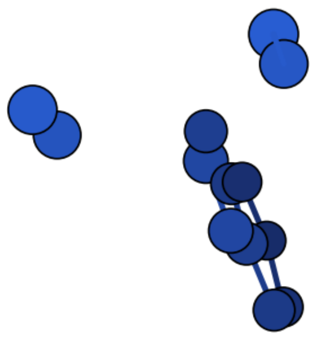




initiale (xTB, N_13^{+}) dernier cycle DFT (50 cycles)

`N13_cation_cation_002` – en cours de fragmentation, 2+2+2+7





initiale (xTB, N_13^{+}) dernier cycle DFT (50 cycles)

`N13_cation_cation_003` – en cours de fragmentation, 2+2+9

Légende générale, Figure S2 : sphères bleues reliées par des bâtonnets = atomes N et liaisons N–N (modèle boules-bâtonnets, rendu depuis les coordonnées cartésiennes .xyz) ; paires groupées par candidat, géométrie initiale GFN2-xTB à gauche et dernière géométrie de la trajectoire d'optimisation DFT (non convergée) à droite, pour les 35 candidats non convergés (Tableau S2).

A.3 Structures avec fréquence imaginaire

Tableau S3. Structures avec au moins une fréquence imaginaire (pas encore de vrai minimum), toutes issues d'un calcul DFT WB97X-D4 terminé. "Max. imaginaire" : la fréquence de plus grande amplitude parmi les modes imaginaires (mode dominant de la coordonnée de réaction vers le vrai minimum). ΔH : énergie relative au ground-state DFT confirmé (vrai minimum, non fragmenté) de la même composition, en kcal/mol – un candidat marqué **Frag.** peut afficher un ΔH négatif : une espèce dissociée (Tableau S1) est souvent plus basse en énergie qu'un vrai cluster lié, sans en être un.

Nom	N	n modes	Max. imag. (cm^{-1})	Intensité (km/mol)	ΔH (kcal/mol)	Frag.
N4_neutral_neutral_002	4	1	-249.3	–	16.85	
N4_neutral_neutral_003	4	1	-567.2	–	33.07	
N5_anion_anion_006	5	1	-211.7	–	104.46	
N5_anion_anion_009	5	1	-15.9	–	66.07	✓
N5_cation_cation_007	5	2	-92.3	–	34.45	✓
N6_neutral_neutral_004	6	3	-26.9	–	-172.76	✓
N6_neutral_neutral_010	6	2	-779.2	–	149.79	
N7_anion_anion_003	7	1	-210.7	–	20.91	
N7_anion_anion_007	7	1	-746.3	–	35.76	
N7_anion_anion_009	7	1	-112.9	–	54.96	
N7_anion_anion_013	7	1	-238.3	–	9.42	
N7_anion_anion_016	7	1	-275.3	–	40.69	
N7_anion_anion_017	7	1	-209.3	–	80.80	
N7_anion_anion_018	7	1	-24.6	–	79.33	
N7_anion_anion_020	7	2	-1284.2	–	179.57	
N7_cation_cation_005	7	2	-697.7	–	189.69	
N7_cation_cation_006	7	1	-297.4	–	200.56	
N8_neutral_neutral_007	8	2	-193.6	–	88.70	
N8_neutral_neutral_018	8	3	-25.0	–	-202.57	✓
N9_anion_anion_006	9	2	-285.6	–	27.56	
N9_anion_anion_010	9	1	-50.8	–	62.59	
N9_anion_anion_015	9	1	-17.7	–	-10.16	✓
N9_cation_cation_008	9	1	-103.5	–	7.40	
N10_neutral_neutral_013	10	2	-255.6	–	193.75	
N11_anion_anion_001	11	2	-66.8	–	-60.32	
N11_anion_anion_002	11	1	-44.4	–	-21.85	
N11_anion_anion_003	11	2	-322.0	–	48.86	
N11_anion_anion_004	11	1	-79.0	–	31.83	
N11_anion_anion_005	11	1	-63.6	–	14.07	
N11_anion_anion_010	11	1	-42.9	–	41.94	
N11_cation_cation_002	11	1	-138.2	–	-23.30	
N11_cation_cation_005	11	1	-248.0	–	22.54	

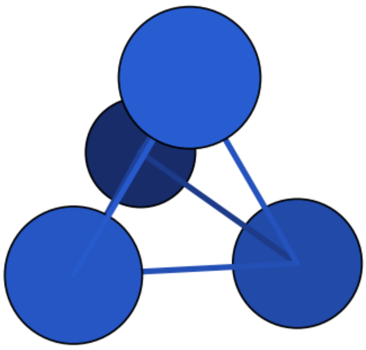
(suite)

N11_cation_cation_009	11	2	-526.8	–	155.19	✓
N11_cation_cation_010	11	2	-526.8	–	155.19	✓
N12_neutral_neutral_005	12	1	-28.0	–	138.12	
N14_neutral_neutral_001	14	1	-81.0	–	4.42	
N14_neutral_neutral_004	14	6	-36.3	–	-246.16	✓
N16_neutral_neutral_001	16	1	-135.1	–	40.23	

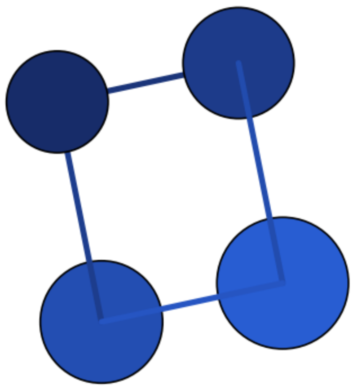
A.4 Structures optimisées confirmées

Les 110 candidats confirmés comme vrais minima **non-fragmentés** au moment de la rédaction, deux par ligne. Trois figures, classées par nombre d'atomes croissant puis par stabilité au sein de chaque formule : Figure S3 (neutres), Figure S4 (cations), Figure S5 (anions). Chaque structure porte son nom, son groupe ponctuel, son ΔH (kcal/mol), le niveau de théorie de ce ΔH (DFT ou xTB si le DFT n'est pas encore terminé), et sa référence (numéro(s) d'article ou *our work*). Comme dans les Tableaux 2–4 : ΔH = énergie d'isomérisation à N et charge fixés, N_x^q(isomère i) $\rightarrow$ N_x^q(isomère le plus stable non-fragmenté de la même formule), $\Delta H = E(i) - E_{\text{stable}}$ (isomère le plus stable $= 0$).

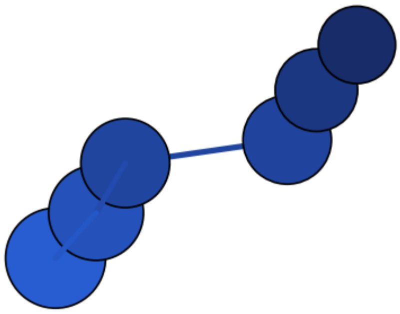
Figure S3 – composés neutres



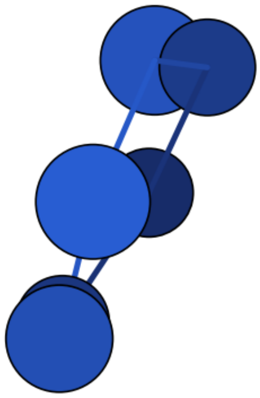
N4_neutral_neutral_001 – Td – $\Delta H = 0.00$ (DFT) – our work



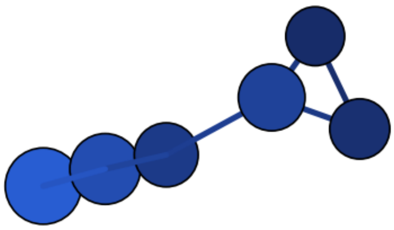
N4_neutral_neutral_004 – D2h – $\Delta H = 3.38$ (DFT) – our work



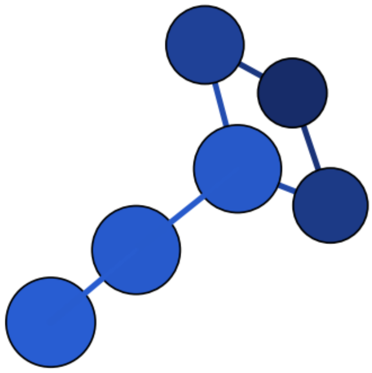
`N6_neutral_neutral_001` – C2 – $\Delta H = 0.00$ (DFT) – our work



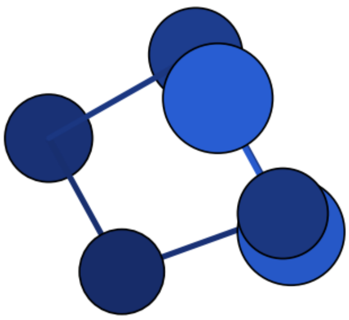
`N6_neutral_neutral_002` – D2 – $\Delta H = 23.07$ (DFT) – [2], our work



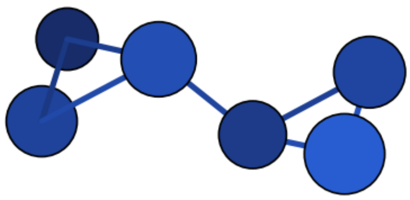
`N6_neutral_neutral_003` – C1 – $\Delta H = 28.07$ (DFT) – our work



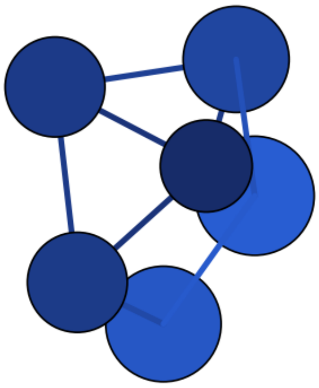
`N6_neutral_neutral_005` – C1 – $\Delta H = 52.69$ (DFT) – our work



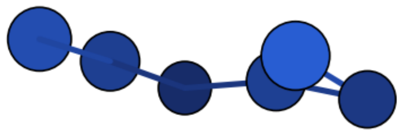
`N6_neutral_neutral_007` – C2 – $\Delta H = 53.36$ (DFT) – our work



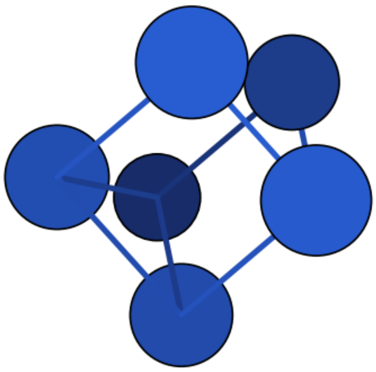
`N6_neutral_neutral_009` – C2 – $\Delta H = 58.25$ (DFT) – our work



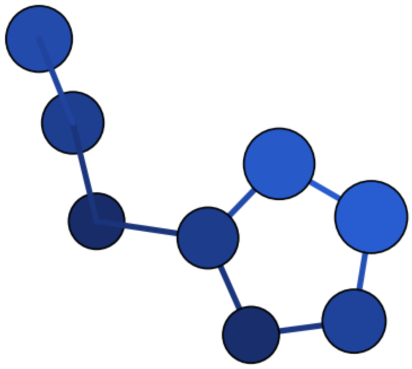
`N6_neutral_neutral_008` – C2v – $\Delta H = 63.90$ (DFT) – our work



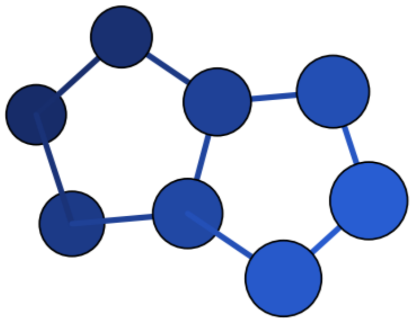
`N6_neutral_neutral_006` – C1 – $\Delta H = 85.49$ (DFT) – our work



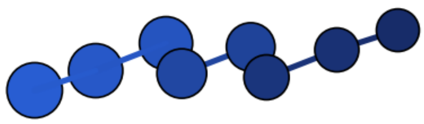
`N6_neutral_neutral_012` – D3h – $\Delta H = 132.46$ (DFT) – our work



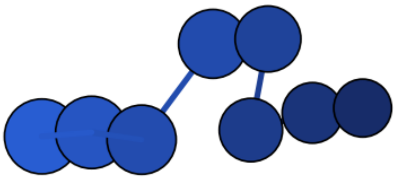
`N8_neutral_neutral_002` – C1 – $\Delta H = 0.00$ (DFT) – [1],[2], our work



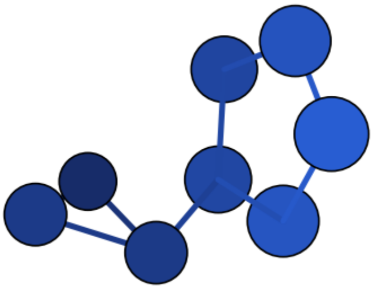
`N8_neutral_neutral_001` – D2h – $\Delta H = 11.23$ (DFT) – [1],[2], our work



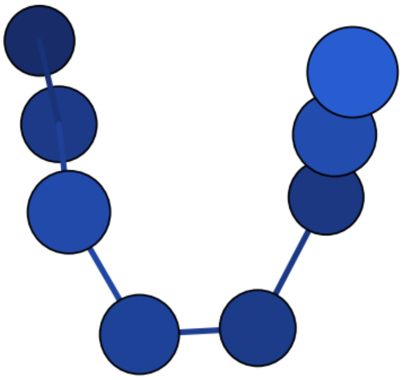
`N8_neutral_neutral_005` – C2 – $\Delta H = 18.41$ (DFT) – [2], our work



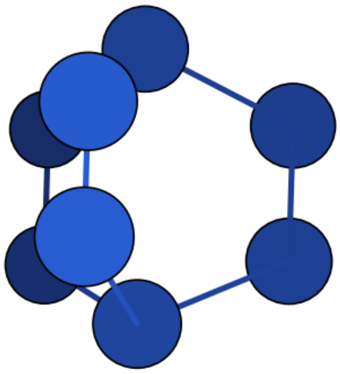
`N8_neutral_neutral_003` – C1 – $\Delta H = 18.53$ (DFT) – [2], our work



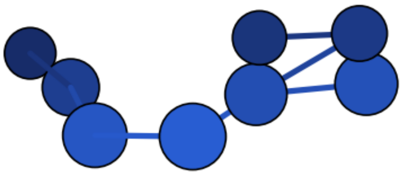
`N8_neutral_neutral_008` – C1 – $\Delta H = 31.11$ (DFT) – our work



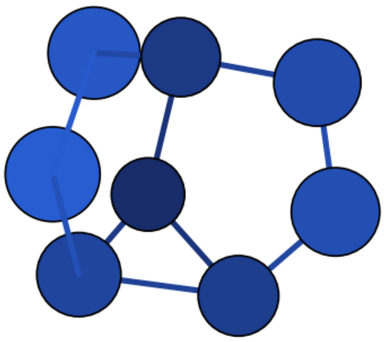
`N8_neutral_neutral_004` – C1 – $\Delta H = 36.64$ (DFT) – [2], our work



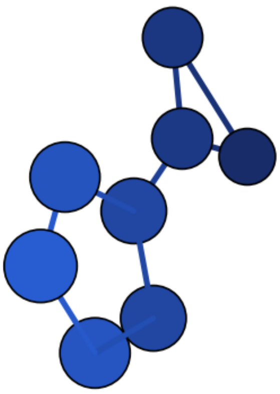
`N8_neutral_neutral_009` – D3h – $\Delta H = 68.56$ (DFT) – our work



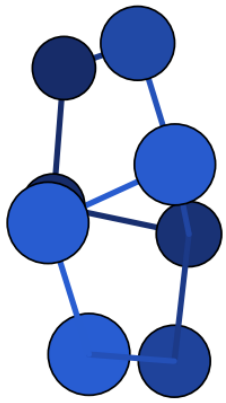
`N8_neutral_neutral_010` – Cs – $\Delta H = 76.55$ (DFT) – our work



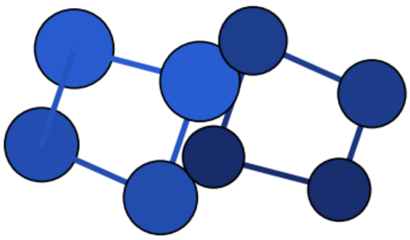
`N8_neutral_neutral_012` – Cs – $\Delta H = 88.83$ (DFT) – our work



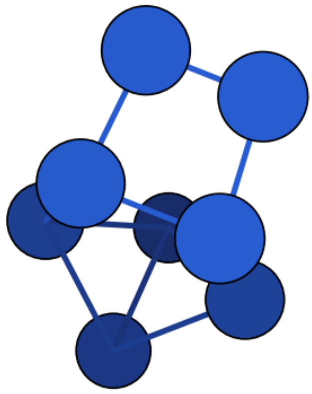
`N8_neutral_neutral_013` – C2 – $\Delta H = 101.43$ (DFT) – our work



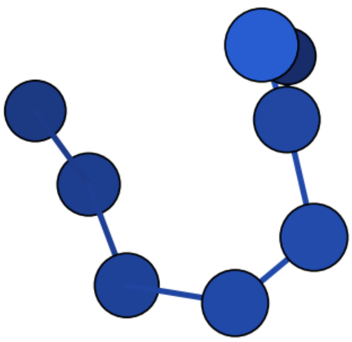
`N8_neutral_neutral_019` – D2d – $\Delta H = 114.22$ (DFT) – our work



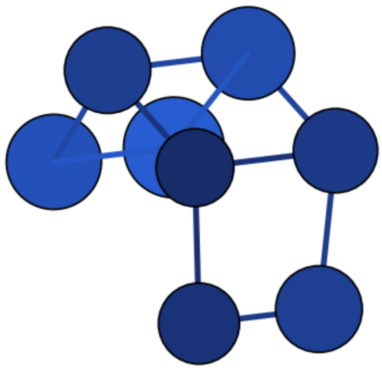
`N8_neutral_neutral_020` – C2h – $\Delta H = 118.58$ (DFT) – our work



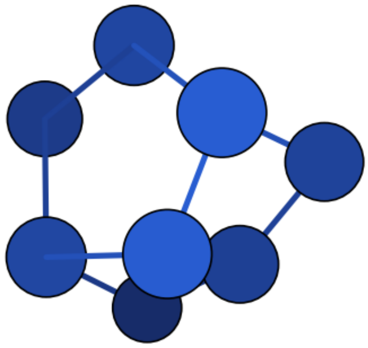
`N8_neutral_neutral_016` – Cs – $\Delta H = 122.30$ (DFT) – our work



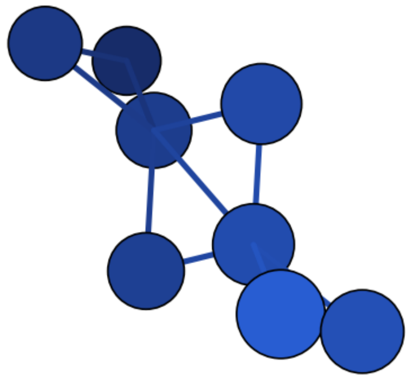
`N8_neutral_neutral_014` – C1 – $\Delta H = 128.12$ (DFT) – our work



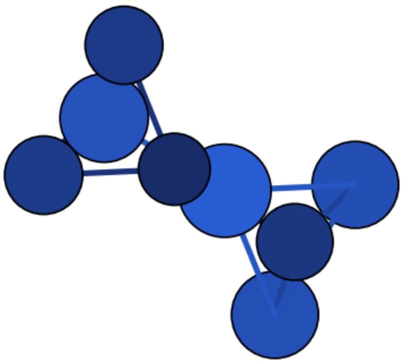
`N8_neutral_neutral_021` – C2 – $\Delta H = 129.00$ (DFT) – our work



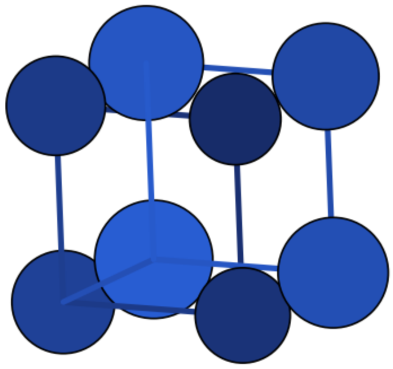
`N8_neutral_neutral_022` – C1 – $\Delta H = 169.70$ (DFT) – our work



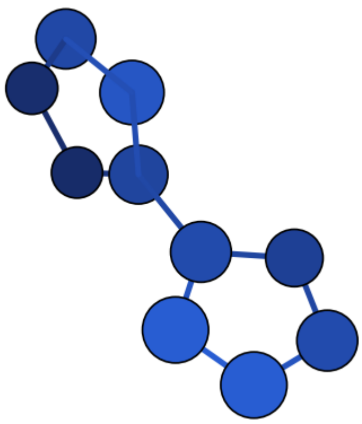
`N8_neutral_neutral_024` – D2h – $\Delta H = 178.42$ (DFT) – our work



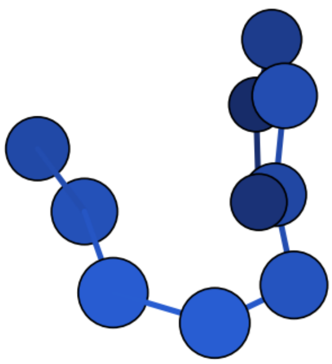
`N8_neutral_neutral_023` – D2h – $\Delta H = 183.12$ (DFT) – our work



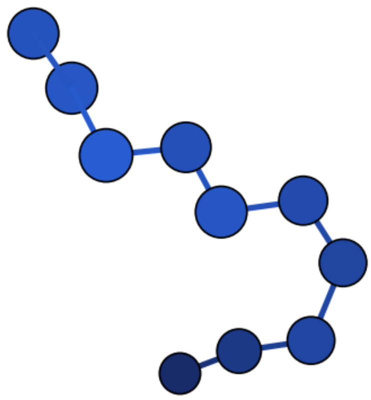
`N8_neutral_neutral_025` – Oh – $\Delta H = 220.10$ (DFT) – our work



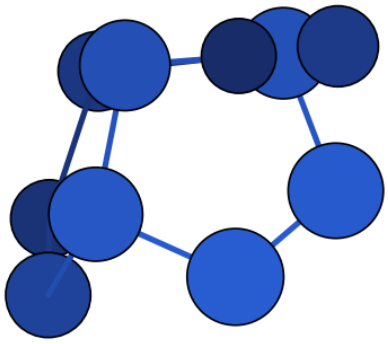
`N10_neutral_neutral_001` – C2 – $\Delta H = 0.00$ (DFT) – our work



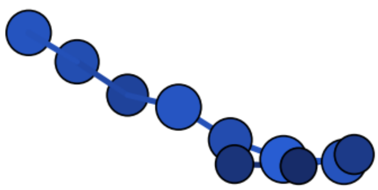
`N10_neutral_neutral_002` – C1 – $\Delta H = 30.61$ (DFT) – our work



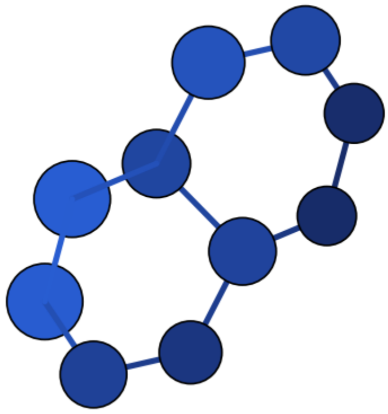
`N10_neutral_neutral_003` – C1 – $\Delta H = 50.60$ (DFT) – [2], our work



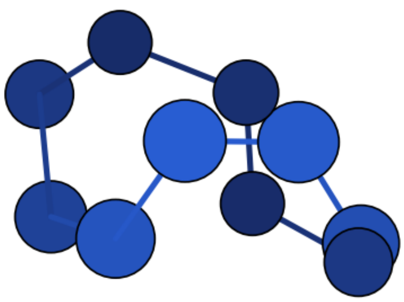
`N10_neutral_neutral_006` – Cs – $\Delta H = 69.56$ (DFT) – our work



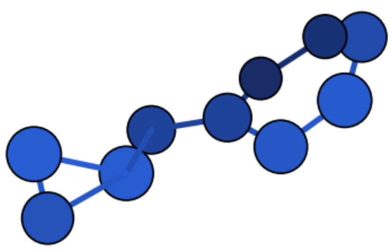
`N10_neutral_neutral_004` – C1 – $\Delta H = 71.13$ (DFT) – [2], our work



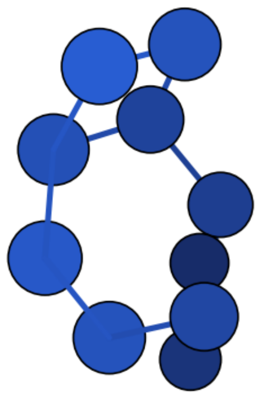
`N10_neutral_neutral_007` – Cs – $\Delta H = 87.34$ (DFT) – our work



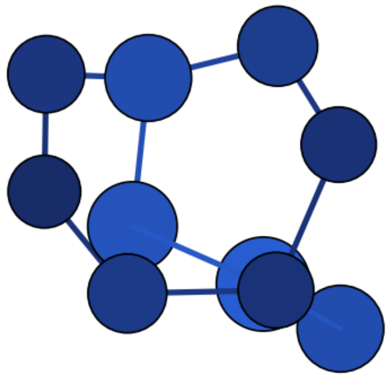
`N10_neutral_neutral_005` – C1 – $\Delta H = 102.77$ (DFT) – our work



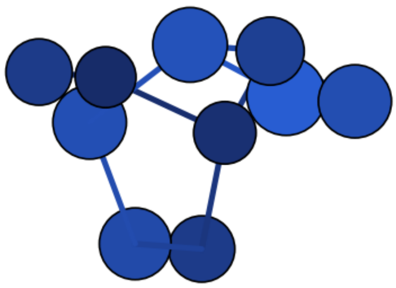
`N10_neutral_neutral_011` – C1 – $\Delta H = 107.23$ (DFT) – our work



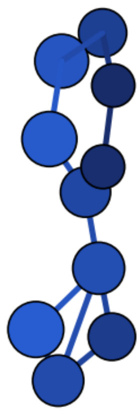
`N10_neutral_neutral_010` – C1 – $\Delta H = 118.99$ (DFT) – our work



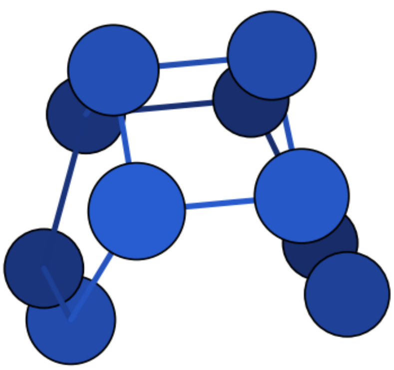
`N10_neutral_neutral_015` – C2 – $\Delta H = 123.40$ (DFT) – our work



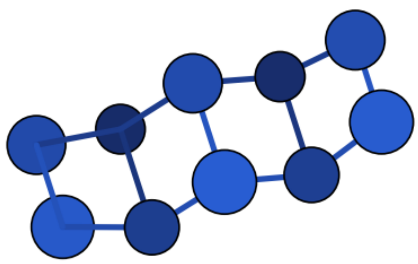
`N10_neutral_neutral_016` – C1 – $\Delta H = 126.63$ (DFT) – our work



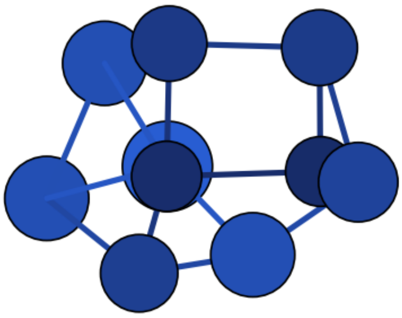
`N10_neutral_neutral_014` – C1 – $\Delta H = 139.29$ (DFT) – our work



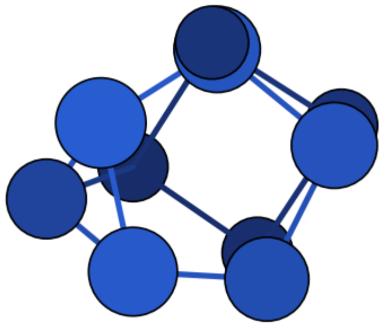
`N10_neutral_neutral_017` – C2v – $\Delta H = 151.14$ (DFT) – our work



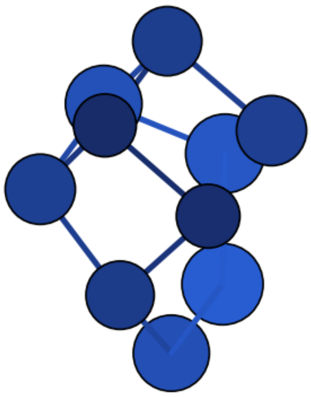
`N10_neutral_neutral_021` – C2v – $\Delta H = 173.84$ (DFT) – our work



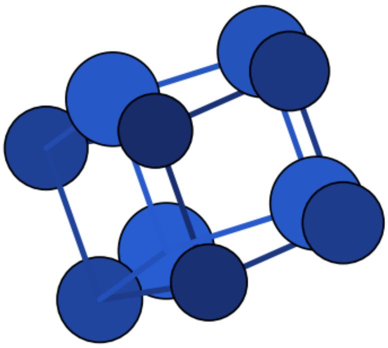
`N10_neutral_neutral_018` – C2 – $\Delta H = 199.07$ (DFT) – our work



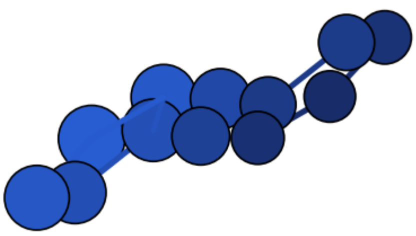
`N10_neutral_neutral_019` – C3v – $\Delta H = 203.80$ (DFT) – our work



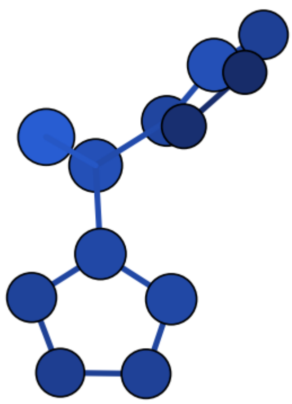
`N10_neutral_neutral_020` – C2 – $\Delta H = 214.21$ (DFT) – our work



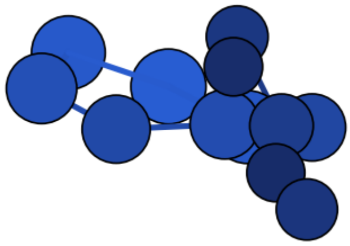
`N10_neutral_neutral_022` – D5h – $\Delta H = 241.14$ (DFT) – our work



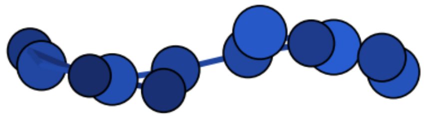
`N12_neutral_neutral_001` – C1 – $\Delta H = 0.00$ (DFT) – our work



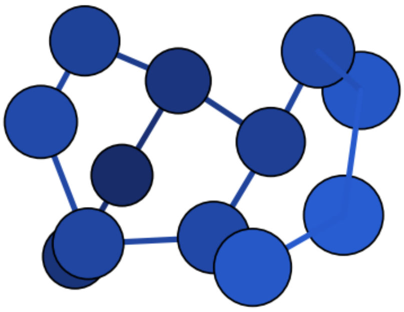
`N12_neutral_neutral_003` – C1 – $\Delta H = 36.41$ (DFT) – our work



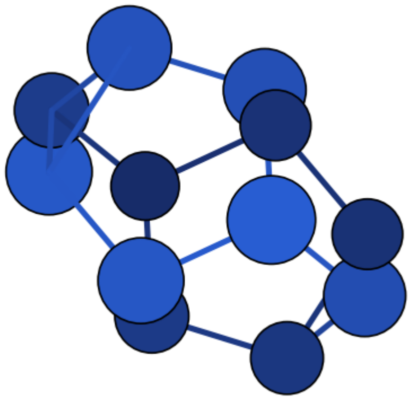
`N12_neutral_neutral_002` – C1 – $\Delta H = 56.99$ (DFT) – our work



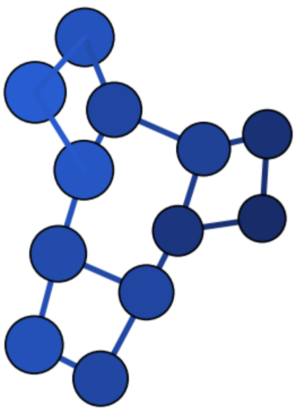
`N12_neutral_neutral_004` – Cs – $\Delta H = 116.12$ (DFT) – our work



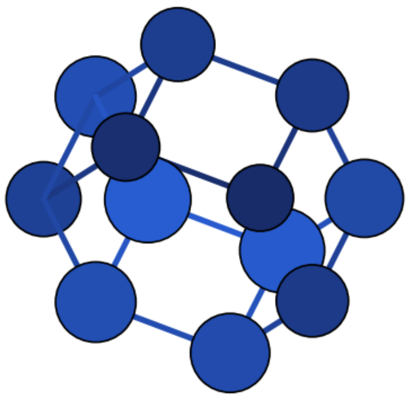
`N12_neutral_neutral_006` – C1 – $\Delta H = 127.50$ (DFT) – our work



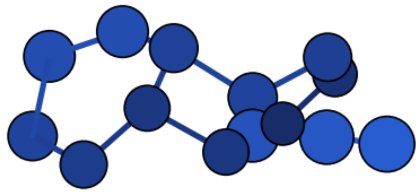
`N12_neutral_neutral_007` – D3d – $\Delta H = 188.33$ (DFT) – our work



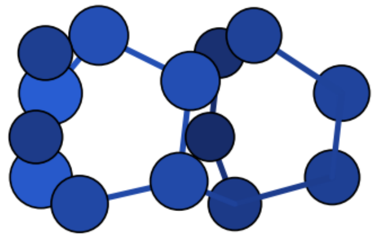
`N12_neutral_neutral_008` – C1 – $\Delta H = 219.30$ (DFT) – our work



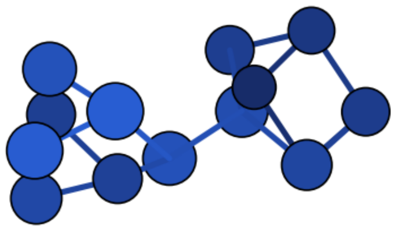
`N12_neutral_neutral_009` – D6h – $\Delta H = 334.92$ (DFT) – our work



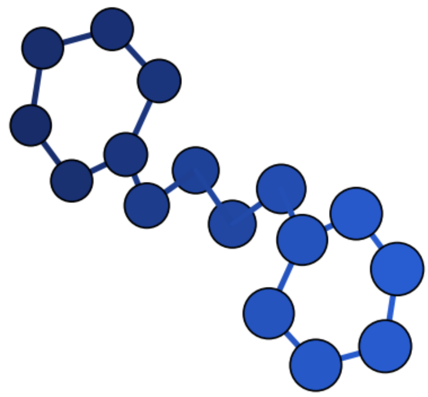
`N14_neutral_neutral_002` – C1 – $\Delta H = 0.00$ (DFT) – our work



`N14_neutral_neutral_003` – C2v – $\Delta H = 22.06$ (DFT) – our work

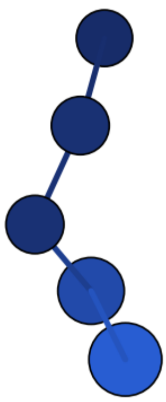


`N14_neutral_neutral_005` – C2 – $\Delta H = 95.19$ (DFT) – our work

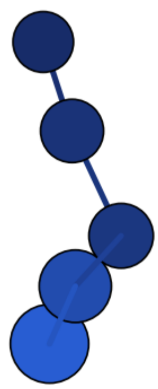


`N16_neutral_neutral_002` – C1 – $\Delta H = 0.00$ (DFT) – our work

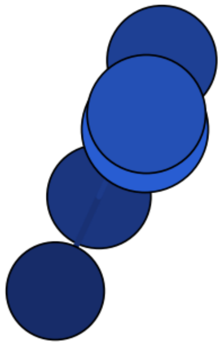
Figure S4 – composés cationiques



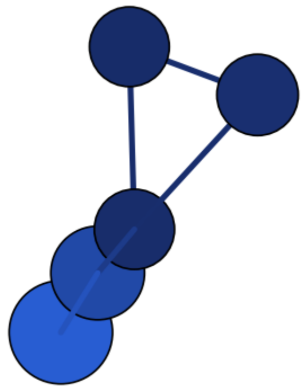
`N5_cation_cation_003` – C1 – $\Delta H = 0.00$ (DFT) – our work



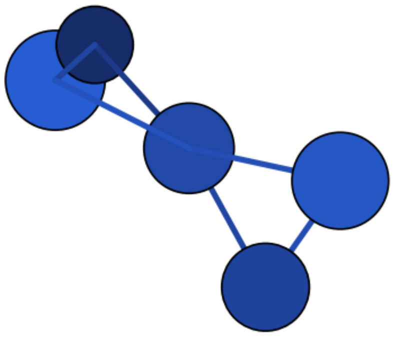
`N5_cation_cation_001` – C2v – $\Delta H = 0.00$ (DFT) – our work



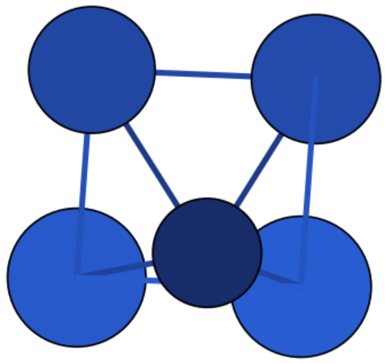
N5_cation_cation_004 – C1 – $\Delta H = 0.01$
(DFT) – our work



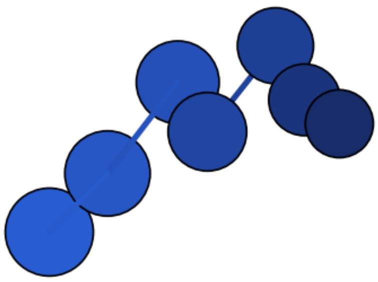
N5_cation_cation_002 – Cs – $\Delta H = 41.63$
(DFT) – our work



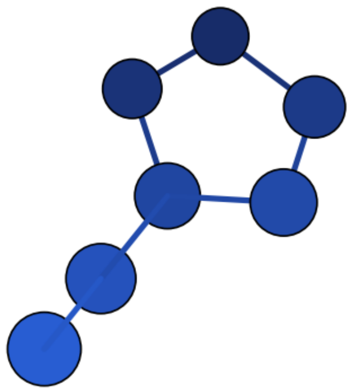
N5_cation_cation_005 – D2d – $\Delta H = 109.53$
(DFT) – our work



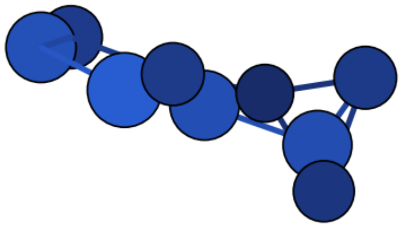
N5_cation_cation_006 – C4v – $\Delta H = 155.74$
(DFT) – our work



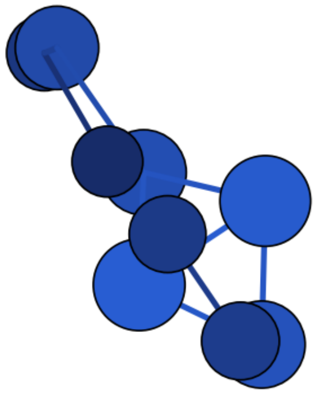
N7_cation_cation_001 – Cs – $\Delta H = 0.00$
(DFT) – our work



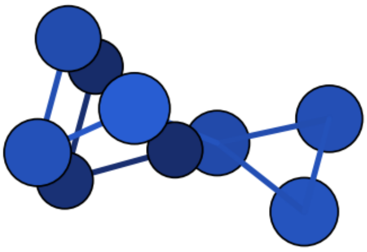
N7_cation_cation_003 – Cs – $\Delta H = 25.42$
(DFT) – our work



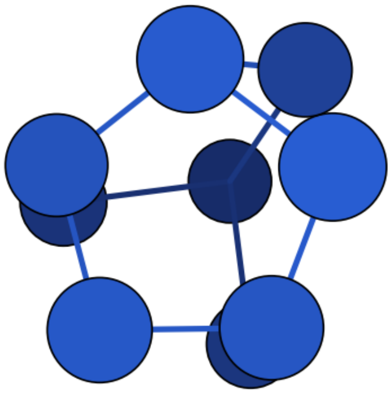
`N9_cation_cation_006` – C2v – $\Delta H = 0.00$ (DFT) – our work



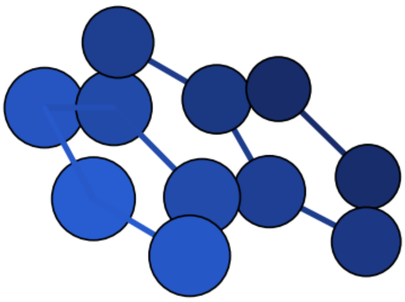
`N9_cation_cation_009` – C1 – $\Delta H = 38.07$ (DFT) – our work



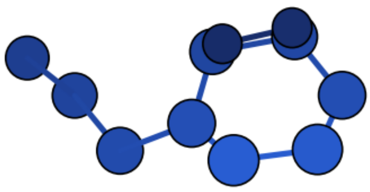
`N9_cation_cation_012` – C2v – $\Delta H = 57.61$ (DFT) – our work



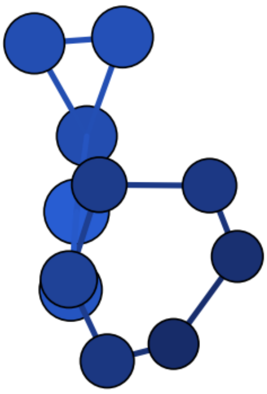
`N9_cation_cation_011` – Cs – $\Delta H = 82.84$ (DFT) – our work



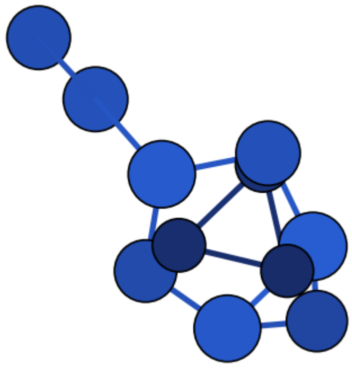
`N11_cation_cation_001` – C2v – $\Delta H = 0.00$ (DFT) – our work



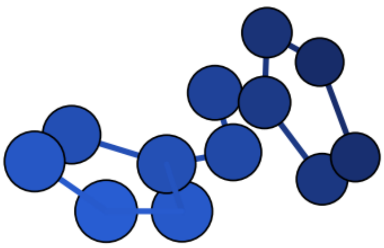
`N11_cation_cation_004` – C1 – $\Delta H = 32.75$ (DFT) – our work



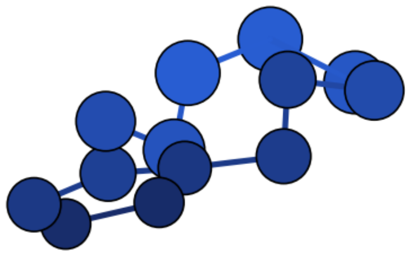
N11_cation_cation_006 – C1 – $\Delta H = 44.38$ (DFT) – our work



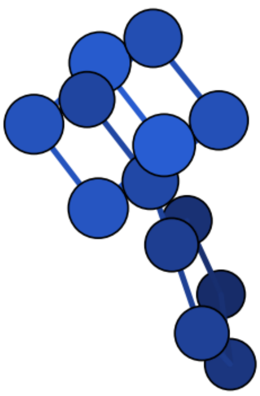
N11_cation_cation_007 – C1 – $\Delta H = 95.39$ (DFT) – our work



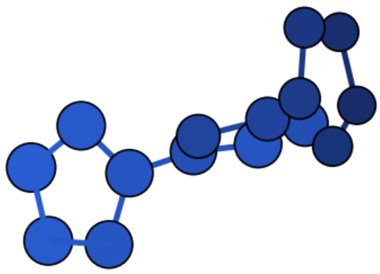
N12_cation_cation_001 – C1 – $\Delta H = 0.00$ (DFT) – our work



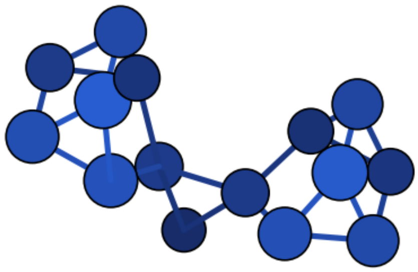
N13_cation_cation_001 – C1 – $\Delta H = 0.00$ (DFT) – our work



N13_cation_cation_004 – C1 – $\Delta H = 151.22$ (DFT) – our work

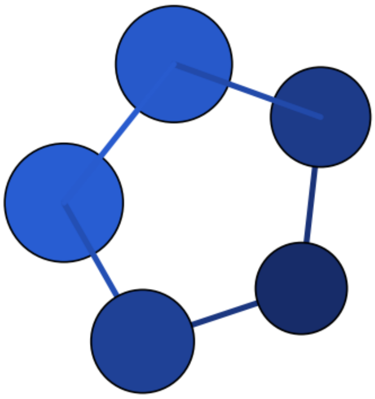


N15_cation_cation_001 – C1 – $\Delta H = 0.00$ (DFT) – our work

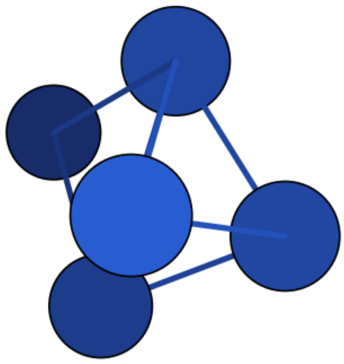


`N15_cation_cation_002` – C1 – $\Delta H = 327.87$ (DFT) – our work

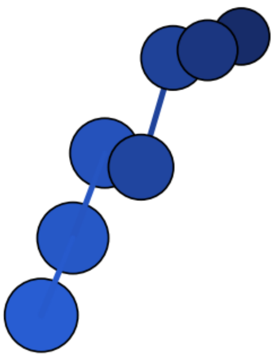
Figure S5 – composés anioniques



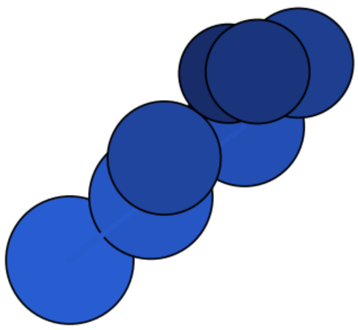
`N5_anion_anion_001` – D5h – $\Delta H = 0.00$ (DFT) – our work



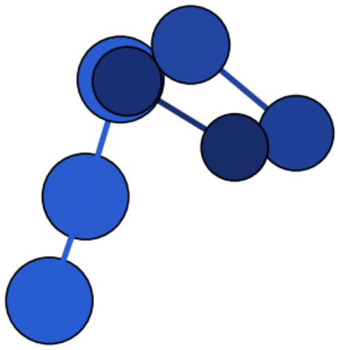
`N5_anion_anion_010` – C2v – $\Delta H = 170.76$ (DFT) – our work



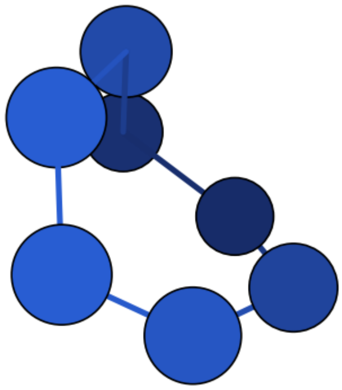
`N7_anion_anion_002` – C1 – $\Delta H = 0.00$ (DFT) – our work



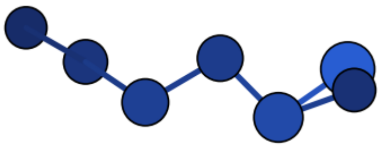
`N7_anion_anion_001` – C1 – $\Delta H = 0.00$ (DFT) – our work



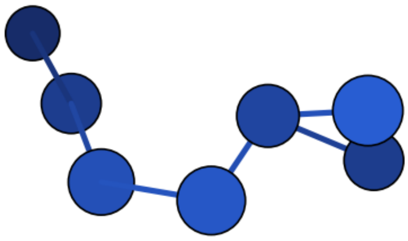
`N7_anion_anion_005` – C1 – $\Delta H = 2.22$ (DFT) – our work



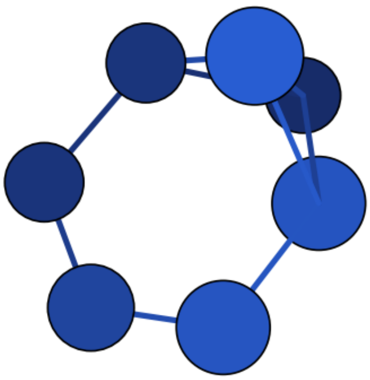
`N7_anion_anion_004` – C1 – $\Delta H = 7.80$ (DFT) – our work



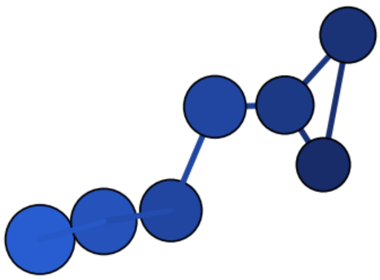
`N7_anion_anion_011` – C1 – $\Delta H = 26.27$ (DFT) – our work



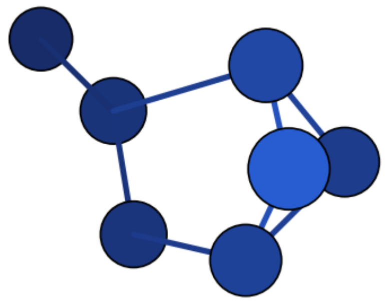
`N7_anion_anion_012` – C1 – $\Delta H = 29.35$ (DFT) – our work



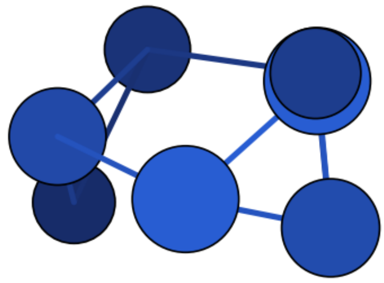
`N7_anion_anion_010` – C2v – $\Delta H = 46.25$ (DFT) – our work



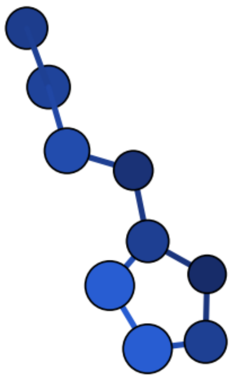
`N7_anion_anion_008` – C1 – $\Delta H = 49.34$ (DFT) – our work



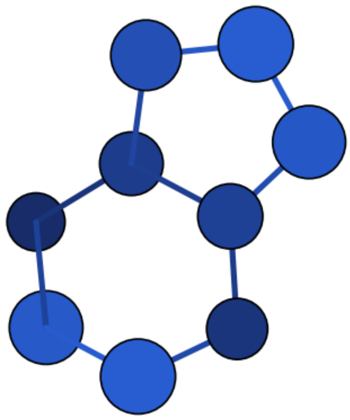
`N7_anion_anion_014` – Cs – $\Delta H = 55.87$ (DFT) – our work



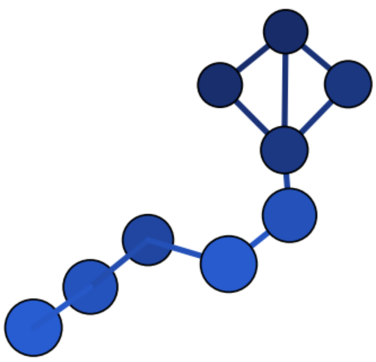
`N7_anion_anion_019` – C1 – $\Delta H = 89.71$ (DFT) – our work



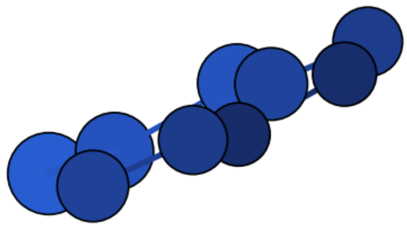
`N9_anion_anion_001` – C1 – $\Delta H = 0.00$ (DFT) – our work



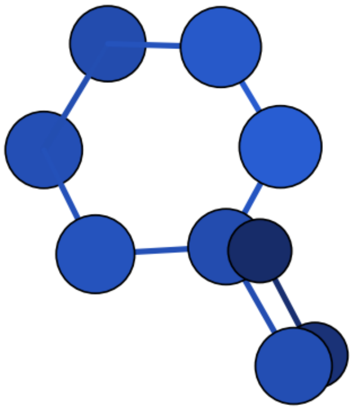
`N9_anion_anion_003` – C1 – $\Delta H = 40.85$ (DFT) – our work



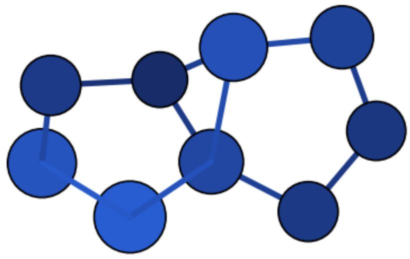
`N9_anion_anion_008` – C1 – $\Delta H = 48.03$ (DFT) – our work



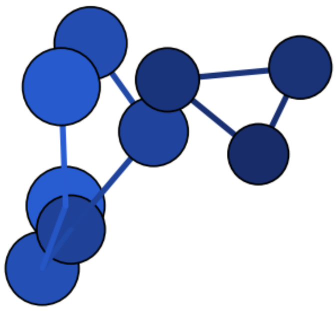
`N9_anion_anion_004` – C1 – $\Delta H = 50.71$ (DFT) – our work



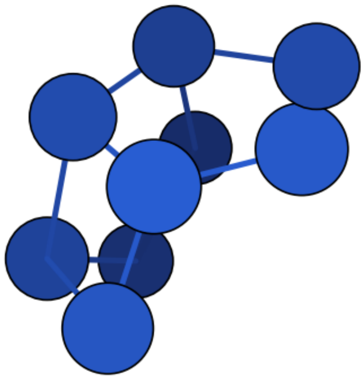
`N9_anion_anion_011` – Cs – $\Delta H = 58.34$ (DFT) – our work



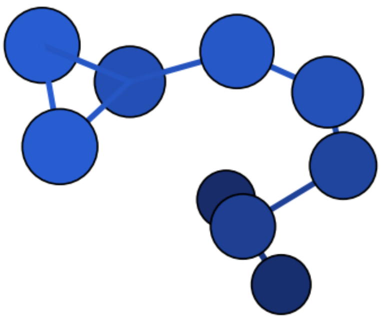
`N9_anion_anion_005` – C2 – $\Delta H = 59.03$ (DFT) – our work



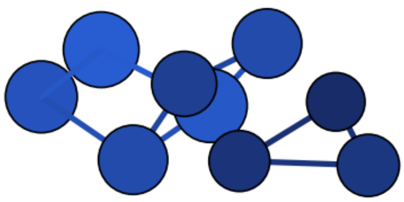
`N9_anion_anion_012` – C1 – $\Delta H = 62.73$ (DFT) – our work



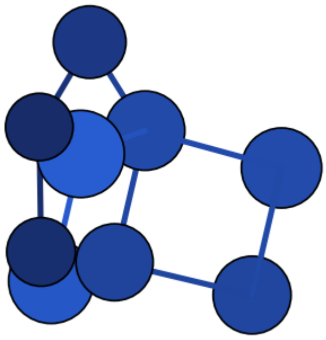
`N9_anion_anion_013` – C1 – $\Delta H = 96.48$ (DFT) – our work



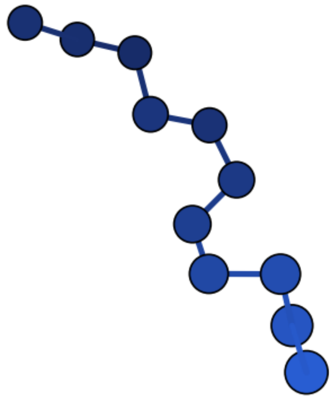
`N9_anion_anion_018` – C1 – $\Delta H = 98.82$ (DFT) – our work



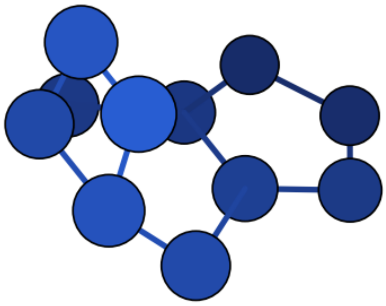
`N9_anion_anion_019` – C1 – $\Delta H = 139.31$ (DFT) – our work



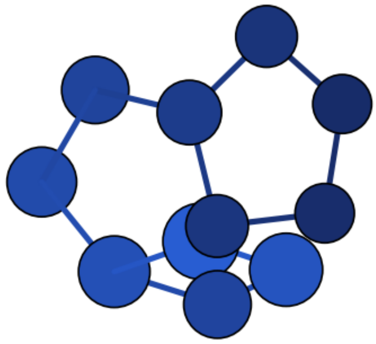
N9_anion_anion_016 – C2v – $\Delta H = 164.79$ (DFT) – our work



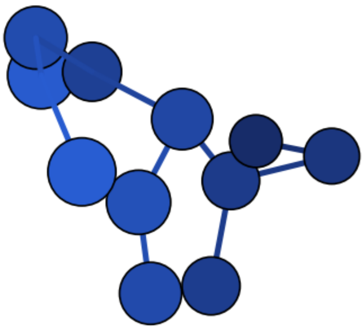
N11_anion_anion_006 – C1 – $\Delta H = 0.00$ (DFT) – our work



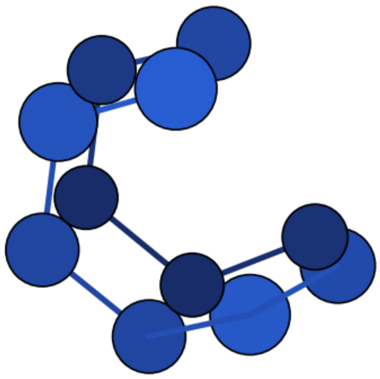
N11_anion_anion_007 – Cs – $\Delta H = 41.41$ (DFT) – our work



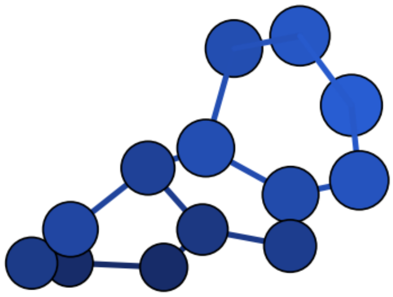
N11_anion_anion_008 – C1 – $\Delta H = 48.28$ (DFT) – our work



N11_anion_anion_013 – C1 – $\Delta H = 116.68$ (DFT) – our work



N11_anion_anion_017 – C1 – $\Delta H = 125.14$ (DFT) – our work



`N13_anion_anion_001` – C1 – $\Delta H = 0.00$
(DFT) – our work

A.5 Références

Articles cités dans les tableaux de la section 2 (numérotation propre à ce rapport ; correspondance avec les codes de `biblio_polyN/table_references.tex` de `polyN-pipeline` en annotation).

Références extraites (structure + ΔH_f isodesmique dans `polyN-pipeline`)

Ces 22 références ont chacune fourni au moins une structure recalculée (GFN2-xTB, isodesmique) dans `polyN-pipeline`. Parmi elles, 2 correspondent en plus à la topologie d'au moins un de nos 193 candidats DFT (repérées *citée dans ce rapport* ; numéro(s) de citation reportés en §2).

1. Glukhovtsev, Jiao, von Ragé Schleyer, *Inorg. Chem.* **1996**, 35, 7124–7133. – *structure extraite, citée dans ce rapport*

2. Fau, Mobita, Wilson, Perera, Bartlett, *Quantum Theory Project report*, University of Florida. – *structure extraite, citée dans ce rapport*

3. Bennett F.R., Maier J.P., Chambaud G., Rosmus P., *Photodissociation, charge and atom transfer processes in electronically excited states of N3+*, Chemical Physics 209 (1996) 275. – *structure extraite*

4. Fau S., Wilson K.J., Bartlett R.J., *On the Stability of N5+N5-*, J. Phys. Chem. A 2002, 106, 4639. – *structure extraite*

5. Leininger M.L., Van Huis T.J., Schaefer H.F. III, *Protonated High Energy Density Materials: N4 Tetrahedron and N8 Octahedron*, J. Phys. Chem. A 1997, 101, 4460. – *structure extraite*

6. Bittererova M., Brinck T., *Theoretical Study of the Triplet N4 Potential Energy Surface*, J. Phys. Chem. A 2000, 104, 11999. – *structure extraite*

7. Gimarc B.M., Zhao M., *Strain Energies in Homoatomic Nitrogen Clusters N4, N6, and N8*, Inorg. Chem. 1996, 35, 3289. – *structure extraite*

8. Glukhovtsev M.N., Laiter S., *Thermochemistry of Tetrazete and Tetraazatetrahedrane: A High-Level Computational Study*, J. Phys. Chem. 1996, 100, 1569. – *structure extraite*

9. Nguyen M.T., Ha T.K., *Decomposition mechanism of the polynitrogen N5 and N6 clusters and their ions*, Chem. Phys. Lett. 335 (2001) 311. – *structure extraite*

10. Vij A., Wilson W.W., Vij V., Tham F.S., Sheehy J.A., Christe K.O., *Polynitrogen Chemistry. Synthesis, Characterization, and Crystal Structure of Surprisingly Stable Fluoroantimonate Salts of N5+*, J. Am. Chem. Soc. 2001, 123, 6308. – *structure extraite*

11. Gagliardi L., Evangelisti S., Barone V., Roos B.O., *On the dissociation of N6 into 3 N2 molecules*, Chem. Phys. Lett. 320 (2000) 518. – *structure extraite*

12. Li Q.S., Hu X.G., Xu W.G., *Structure and stability of N7 cluster*, Chem. Phys. Lett. 287 (1998) 94. – *structure extraite*

13. Liu Y.D., Zhao J.F., Li Q.S., *Structures and stability of N7+ and N7- clusters*, Theor. Chem. Acc. 107 (2002) 140. – *structure extraite*

14. Ha T.K., Suleimenov O., Nguyen M.T., *A quantum chemical study of three isomers of N20*, Chem. Phys. Lett. 315 (1999) 327. – *structure extraite*

15. Li Q.S., Wang L.J., *Theoretical Studies on the Potential Energy Surfaces of N8 Clusters*, J. Phys. Chem. A (recv. Aug 2000). – *structure extraite*

16. Chung G., Schmidt M.W., Gordon M.S., *An Ab Initio Study of Potential Energy Surfaces for N8 Isomers*, J. Phys. Chem. A 2000, 104, 5647. – *structure extraite*

17. Gagliardi L., Evangelisti S., Roos B.O., Widmark P.O., *A theoretical study of ten N8 isomers*, J. Mol. Struct. (Theochem) 428 (1998) 1-8. – *structure extraite*

18. Wang L.J., Warburton P., Mezey P.G., *Theoretical Prediction on the Synthesis Reaction Pathway of N6 (C2h)*, J. Phys. Chem. A 2002, 106, 2748. – *structure extraite*

19. Wang X., Ren Y., Shuai M.B., Wong N.B., Li W.K., Tian A.M., *Structure and stability of new N7 isomers*, J. Mol. Struct. (Theochem) 538 (2001) 145. – *structure extraite*

20. Thompson M.D., Bledson T.M., Strout D.L., *Dissociation Barriers for Odd-Numbered Acyclic Nitrogen Molecules N9 and N11*, J. Phys. Chem. A 2002. – *structure extraite*

21. Gu J.D., Chen K.X., Jiang H.L., Chen J.Z., Ji R.Y., Ren Y., Tian A.M., *N18: a computational investigation*, J. Mol. Struct. (Theochem) 428 (1998) 183. – *structure extraite*

22. Klapotke T.M., Harcourt R.D., *The interconversion of N12 to N8 and two equivalents of N2*, J. Mol. Struct. (Theochem) 541 (2001) 237. – *structure extraite*

Corpus de recherche bibliographique (OpenAlex), structures non extraites – travail restant

156 articles identifiés par recherche de citations (générations 1 et 2, articles citant les 39 sources de `polyN-pipeline`, filtrés aux composés exclusivement azotés) – **aucun n'a encore de structure extraite ni de ΔH_f recalculé** dans `polyN-pipeline` ; aucun ne peut donc être cité par candidat pour l'instant. **Chevauchement avec la liste précédente vérifié** (correspondance exacte de titre) : 20 des 22 références extraites (A1–A20) sont elles-mêmes parmi les 39 sources dont ce corpus recense les citations – exclues par construction de la liste des 156 citants (un article ne se cite pas lui-même) ; les 2 restantes ([GS] et [B]) n'appartiennent pas non plus à ce corpus (ni comme source, ni comme citant trouvé). **Chevauchement confirmé nul** : les deux listes sont disjointes.

Tableau 14: Travail bibliographique restant à faire (statut : aucune structure extraite, cf. texte ci-dessus) – article, année, DOI.

N°	Année	Titre	DOI
23	1996	Besides N 2 , What Is the Most Stable Molecule Composed Only of Nitrogen Atoms?	`10.1021/ic9606237`
24	1997	High-level computational study on the thermochemistry of saturated and unsaturated three- and four-membered nitrogen and phosphorus rings	`10.1002/(sici)1097-461x(1997)62:4<373::aid-qua5>3.0.co;2-t`
25	1997	New Isomers of N8 without Double Bonds	`10.1021/jp962878b`
26	1999	Homopolyatomare Stickstoffverbindungen	`10.1002/(sici)1521-3757(19990903)111:17<2694::aid-ange2694>3.0.co;2-0`
27	1999	N5+: ein neuartiges homoleptisches Polystickstoff-Ion als Substanz mit hoher Energiedichte	`10.1002/(sici)1521-3757(19990712)111:13/14<2112::aid-ange2112>3.0.co;2-i`
28	1999	Time-of-flight mass spectroscopic detection of new elemental and mixed small atomic clusters in the laser evaporation of carbon nitride	`10.1070/mc1999v009n01abeh001019`
29	2000	An isomeric study of N5+, N5, and N5-: a Gaussian-3 investigation	`10.1016/s0009-2614(00)01071-x`

(suite)

N°	Année	Titre	DOI
30	2000	Dissociation reaction of N8 azapentalene to 4N2: A theoretical study	10.1002/(sici)1097-461x(2000)77:1<311::aid-qua29>3.0.co;2-l
31	2000	Tetrazete (N4). Can it be prepared and observed?	10.1016/s0009-2614(00)00926-x
32	2000	Tetrazete (N4):can it be prepared and observed?	(DOI absent)
33	2000	Theoretical study of the pentanitrogen cation (N5+)	10.1016/s0009-2614(99)01320-2
34	2001	A theoretical study of the nitrogen clusters formed from the ions N3-, N5+, and N5-	10.1063/1.1370063
35	2001	On the nature of bonding in N5+ ion	10.1016/s0166-1280(01)00404-3
36	2001	On the possibility of detecting tetraazatetrahedrane (N4) in liquid or solid nitrogen by Fourier transform Raman spectroscopy	10.1002/jrs.686
37	2001	Structure and Stability of N6 Isomers and Their Spectroscopic Characteristics	10.1021/jp003971+
38	2001	Theoretical study of the N10 clusters without double bonds	10.1002/1097-461x(2001)82:1<34::aid-qua1013>3.0.co;2-1
39	2002	A theoretical study of the azide (N3) doublet states. A new route to tetraazatetrahedrane (N4): N+N3N4	10.1063/1.1476310
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41	2002	ChemInform Abstract: On the Stability of N5+N5.	10.1002/chin.200229002
42	2002	Experimental Detection of Tetranitrogen	10.1126/science.1067681
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45	2002	Synthesis reaction pathway of nitrogen-rich ionic compound N7H2+	10.1016/s0009-2614(02)01042-4
46	2002	Theoretical Study of Potential Energy Surfaces for N12 Clusters	10.1021/jp020110n
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48	2003	A theoretical study on the stability of N15+ cluster	10.1039/b209547e
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50	2003	Density functional calculation of structure and stability of nitrogen clusters N10, N12, and N20	10.1016/s0166-1280(02)00695-4
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56	2003	Structures and kinetic stability of N7 cluster	10.1016/s0009-2614(02)01806-7
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59	2003	THEORETICAL PREDICTION OF POTENTIAL ENERGY SURFACE FOR N 14 CLUSTER	10.1142/s0219633603000331
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66	2004	HighEnergyDensity Materials: Synthesis and Characterization of N5+[P(N3)6]-, N5+ [B(N3)4]-, N5+ [HF2]-n HF, N5+ [BF4]-, N5+ [PF6]-, and N5+ [SO3F]-	10.1002/ange.200454242
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68	2004	Trends in Stability for N18 Cages	10.1021/jp0481153
69	2005	A Gaussian-3 investigation on the stabilities and bonding of the nine N10 clusters	10.1016/ j.theochem.2005.05.035
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74	2005	The race for the first generation of the pentazolate anion in solution is far from over	10.1039/b417010e
75	2006	Computational Studies on Polynitrohexaazaadmantanes as Potential High Energy Density Materials	10.1021/jp0575557
76	2006	Electronic Structure Analysis and Electron Detachment Energies of Polynitrogen Pentagonal Aromatic Anions	10.1021/jp0632743
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78	2007	Computational Aspects of Nitrogen-Rich HEDMs	10.1007/430_2006_053
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82	2007	Nitrogen-Rich Heterocycles	10.1007/430_2006_055
83	2007	Synthesis and Characterization of (Z)[N3NFO]+ and (E)[N3NFO]+	10.1002/anie.200700130
84	2007	The simplest all-nitrogen ring: Photolytically filling the cyclic-N3 well	10.1063/1.2433723
85	2007	Theoretical investigation on the cylinder-shaped N84 cage	10.1016/j.cplett.2007.10.076
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89	2009	Modeling of synthesis and dissociation of the N4 nitrogen cluster of D2H symmetry	10.1007/s11182-010-9362-9
90	2009	Theoretical Study on the Reactions of the Cyclic Trinitrogen Radical toward Oxygen and Water	10.1021/jp810741v
91	2010	Searching for ways to create energetic materials based on polynitrogen compounds (review)	10.1007/s10573-010-0020-x
92	2011	Ab Initio Based Double-Sheeted DMBE Potential Energy Surface for N3(2A") and Exploratory Dynamics Calculations	10.1021/jp2073396
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94	2012	Computational studies on the heats of formation, energetic properties, and thermal stability of energetic nitrogen-rich furazano[3,4-b]pyrazine-based derivatives	10.1016/j.comptc.2012.05.013
95	2012	Density Functional Theoretical Study of Polynitrogen Compounds N5+Y- (Y=B(CF3)4, BF4, PF6 and B(N3)4)	10.1002/cjoc.201280011
96	2012	Theoretical studies on the structures, heats of formation, energetic properties and pyrolysis mechanisms of nitrogen-rich difurazano[3,4-b:3',4'-e]piperazine derivatives and their analogues	10.1007/s11224-012-0133-9
97	2013	Global ab initio ground-state potential energy surface of N4	10.1063/1.4811653
98	2013	Quantum Chemical Study of Aminonitrocyclopentanes as Possible High Energy Density Materials (HEDMs)	(DOI absent)
99	2013	Thermal stability of the N10 compound with extended nitrogen chain	10.1039/c3ra42439a
100	2014	Theoretical study on a novel high-energy density material 4,6,10,12-tetranitro-5,11-bis(nitroimino)-2,8-dioxa-4,6,10,12-tetraaza-tricyclo[7,3,0,03,7]dodecane	10.1039/c3ra46107f
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102	2016	Isomerization Pathways of ONCNO: Unstable or Metastable?	10.1021/acs.jpca.5b12275
103	2016	Polynigrogen Energetic Materials	10.9766/kimst.2016.19.3.319
104	2017	Novel rubidium poly-nitrogen materials at high pressure	10.1063/1.5004416

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106	2017	Smallest fullerene-like clusters in two-probe device junctions: first principle study	10.1080/00268976.2017.1314033
107	2017	Super high-energy density single-bonded trigonal nitrogen allotrope—a chemical twin of the cubic gauche form of nitrogen	10.1039/c6cp08723j
108	2017	The crystal density of nitrogen cubane and other polynitrogen species	10.1007/s00894-017-3454-1
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110	2018	A genetic algorithm survey on closed-shell atomic nitrogen clusters employing a quantum chemical approach	10.1007/s00894-018-3724-6
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112	2018	Screening for potential energetic C-N cages with high energy and good stability: a theoretical comparative study	10.1080/08927022.2018.1540870
113	2018	To Be or Not To Be Protonated: cyclo-N5- in Crystal and Solvent	10.1021/acs.jpclett.8b02841
114	2019	A New Genetic Algorithm Approach Applied to Atomic and Molecular Cluster Studies	10.3389/fchem.2019.00707
115	2019	Cage-like N_106- salt with N-N single bonds	10.1209/0295-5075/124/67004
116	2019	Contemplation on Some Prismatic Polynitrogen Structures - A DFT Treatment	10.1002/zaac.201900131
117	2019	Cycloaddition of molecular dinitrogens: formation of tetrazete anion (N4-; D2h) through associative electron attachment	10.1080/00268976.2019.1607599
118	2019	Structures, Stability, and Decomposition Dynamics of the Polynitrogen Molecule N 5 + B(N 3) 4 - and Its Dimer [N 5 +] 2 [B(N 3) 4 -] 2	10.1021/acs.jpca.9b03704
119	2020	All-Nitrogen Cages and Molecular Crystals: Topological Rules, Stability, and Pyrolysis Paths	10.3390/computation8040091
120	2020	Bipentazole (N10): A Low-Energy Molecular Nitrogen Allotrope with High Intrinsic Stability	10.1021/acs.jpclett.0c01542
121	2020	DFT and CIS (D) theoretical study on the ground and excited states of pentanitrogen cation	10.1002/qua.26393
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123	2020	Frontispiz: Nucleophiler Angriff von Azid auf elektrophile Azide: Bildung von N6Einheiten in Hexazen und Aminopentazolderivaten	10.1002/ange.202083062
124	2020	Many-Body Permutationally Invariant Polynomial Neural Network Potential Energy Surface for N4	10.1021/acs.jctc.0c00430
125	2020	Mixed cationic clusters of nitrogen and hydrogen	10.1063/1.5140850
126	2020	Molecular structures and thermodynamics of stable N4, N6 and N8 neutral poly-nitrogens according to data of QCISD(T)/TZVP method	10.1016/j.cplett.2020.137594
127	2020	Nucleophiler Angriff von Azid auf elektrophile Azide: Bildung von N6Einheiten in Hexazen und Aminopentazolderivaten	10.1002/ange.202003010
128	2020	Properties and Promise of Catenated Nitrogen Systems As High-Energy-Density Materials	10.1021/acs.chemrev.9b00804
129	2020	Stabilization of small nitrogen clusters via spatial constraint	10.1088/1742-6596/1435/1/012062
130	2020	Tetra-, hexa-, and octanitrogen molecules: a quantum chemical design and thermodynamic properties	10.1007/s11172-020-3001-6
131	2021	Accessing the applicability of the MBE approach for constructing potential energy surfaces of nitrogen clusters	10.1016/j.chemphys.2021.111272
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133	2021	Cis-Trans Isomerization and Thermal Decomposition Mechanisms of a Series of Nx (x = 4, 8, 10, 11) Chain-Catenated Energetic Crystals	10.1021/acs.jpca.0c11432
134	2021	From mono-rings to bridged bi-rings to caged bi-rings: a promising design strategy for all-nitrogen high-energy-density materials N10 and N12	10.1039/d1nj00522g
135	2021	How stable can the pentanitrogen cation be in kinetics?	10.1039/d1cc01250a
136	2021	Kinetic Stability of Pentazole	10.1021/acs.jpca.1c06252
137	2021	Pentazoles	10.1016/b978-0-12-818655-8.00123-2
138	2021	Stability of neutral molecular polynitrogens: energy content and decomposition mechanisms	10.1039/d1ra03259c
139	2021	Stabilization mechanisms of three novel full-nitrogen molecules	10.1007/s00706-021-02755-1
140	2021	Theoretical study on the two novel planar-type all-nitrogen N44- anions: Structures, stability, reaction rate and their stable mechanisms via protonation	10.1016/j.cplett.2021.138519
141	2022	A novel all-nitrogen molecular crystal N 16 as a promising high-energy-density material	10.1039/d2dt00820c
142	2022	Conferring all-nitrogen aromatics extra stability by acidic trapping	10.1016/j.molliq.2022.118939

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148	2022	Nitro-based and nitro-free tri-cationic azole salts: a unique class of energetic green tri-ionic salts obtained from the reaction with nitrogen-rich bases	10.1039/d2ma00406b
149	2022	Novel All-Nitrogen Molecular Crystals Composed of Tetragonal N4 Molecules	10.3390/ijms23105503
150	2022	Theoretical design of nitrogen-rich cages for energetic materials	10.1016/j.comptc.2022.113659
151	2023	Bond-alternated and bond-equalized hexazine derivatives	10.1039/d3cp05546a
152	2023	Novel cage-like all-nitrogen molecular crystal stable under ambient conditions	10.1016/j.vacuum.2023.112909
153	2023	Polynitrogen clusters interaction with water: experimental and theoretical perspectives	10.1088/1361-6463/acec84
154	2023	Structure and Optical Properties of High-Energy Nitrogen Clusters and Dimers on their Basis	10.1007/s11182-023-02879-3
155	2023	The structural, electronic properties, pressure response and decomposition mechanism of bispentazole (N10) with first principles calculations	10.1016/j.rinp.2023.106743
156	2024	Creating Cyclo-N5+ Cations and Assembling N5+N5- Salts via Electronegativity Comatching in Tailored Ionic Compounds	10.1021/acs.jpcc.4c04367
157	2024	Hexanitrogen (N6): A Synthetic Leap Towards Neutral Nitrogen Allotropes	10.26434/chemrxiv-2024-90vvx
158	2024	Intramolecular steric and relaxation analysis of nitrogen clusters	10.1016/j.comptc.2024.114638
159	2024	Predicting molecular crystals of polynitrogen (N6) structures with cage-like geometries using ab initio evolutionary algorithm	10.1016/j.cplett.2024.141262
160	2024	Quantum tunneling instability of the mythical hexazine and pentazine	10.1039/d3cc05840a
161	2024	Taming cyclo-pentazolate anions with a hydrogen-bonded organic framework	10.1038/s43246-024-00467-7
162	2025	A careful scrutiny of the aromaticity in anionic polynitrogen clusters	10.1039/d5cp01974e
163	2025	Breakthrough in Polynitrogen Chemistry*	10.1002/prep.70084
164	2025	Computational Prediction of a Stable All-Nitrogen Molecular Crystal N 10	10.2139/ssrn.5274458
165	2025	Computational investigation of the [N7]3- anion and related full-nitrogen ring systems	10.1007/s00214-025-03250-0
166	2025	Computational prediction of a stable all-nitrogen molecular crystal N10	10.1063/5.0286764
167	2025	CycloN9-: a Novel 5/6 Fused Polynitrogen Anion for High Energy Density Materials	10.1002/advs.202414394
168	2025	Dual-aromaticity in nitrogen-rich compounds: from fundamental concepts to the application of high-energy-density materials	10.1016/j.ccr.2025.217081
169	2025	Preparation of a neutral nitrogen allotrope hexanitrogen C2h-N6	10.1038/s41586-025-09032-9
170	2025	Theoretical Evaluation of the Energetic Properties of ChainLike Hexanitrogen*	10.1002/prep.70045
171	2025	Theoretical prediction of pressure-stabilized all-nitrogen N 12 molecular crystals with - stacking	10.1039/d5cp01409c
172	2025	Theoretical studies on the long nitrogen chain structure and the constructing mechanism of a catenated N11 cation	10.1139/cjc-2024-0155
173	2026	Energetic Processing of Solid Nitrogen: Infrared Identification of N 4 + and Correlated -Line Emission under Electron Impact and Extreme-UV Irradiation	10.1021/acs.jpca.6c02129
174	2026	High-Pressure Synthesis of K 4 N 6 with All Nitrogen Atoms Forming Aromatic Hexazine [N 6] 4- Anions	10.1021/jacs.5c21993
175	2026	High-Pressure Synthesis of K ₄ N ₆ with All Nitrogen Atoms Forming Aromatic Hexazine [N ₆] ⁴⁻ Anions	10.1021/jacs.5c21993
176	2026	Rational design of nitrogen-rich cage energetic molecules: Balancing energy and stability through controlled nitrogen connectivity	10.1016/j.fuel.2026.139544
177	2026	The quest for ever more energetic materials	10.1039/d6ta04463h
178	2026	The structure of symmetric azoazoles according to ab initio computations	10.1007/s11224-025-02688-z